

Benchmark Comparison Report

Benchmark A vs Benchmark B

This report summarizes the saved experimental results used to compare Benchmark A and Benchmark B across the evaluated image processing and classification stages. Classification results use the Test set, while upstream image processing results use Validation results.

Overall Stage Winner

Benchmark A

Benchmark A Points	Benchmark B Points	Metrics Compared
24.0	11.0	35

Report Contents

- Benchmark algorithm configuration
- Overall prediction performance
- Stage comparison scorecard
- Individual stage descriptions and result tables
- Stage winner and strongest primary metric
- Performance charts and efficiency charts
- Fruit type and ripeness confusion matrices
- Class support distributions
- Radar charts and stage winner distribution
- Final Hybrid Pipeline selection summary

Benchmark Algorithm Configuration

This table lists the exact algorithms and locked parameters used in Benchmark A and Benchmark B for every major evaluated stage.

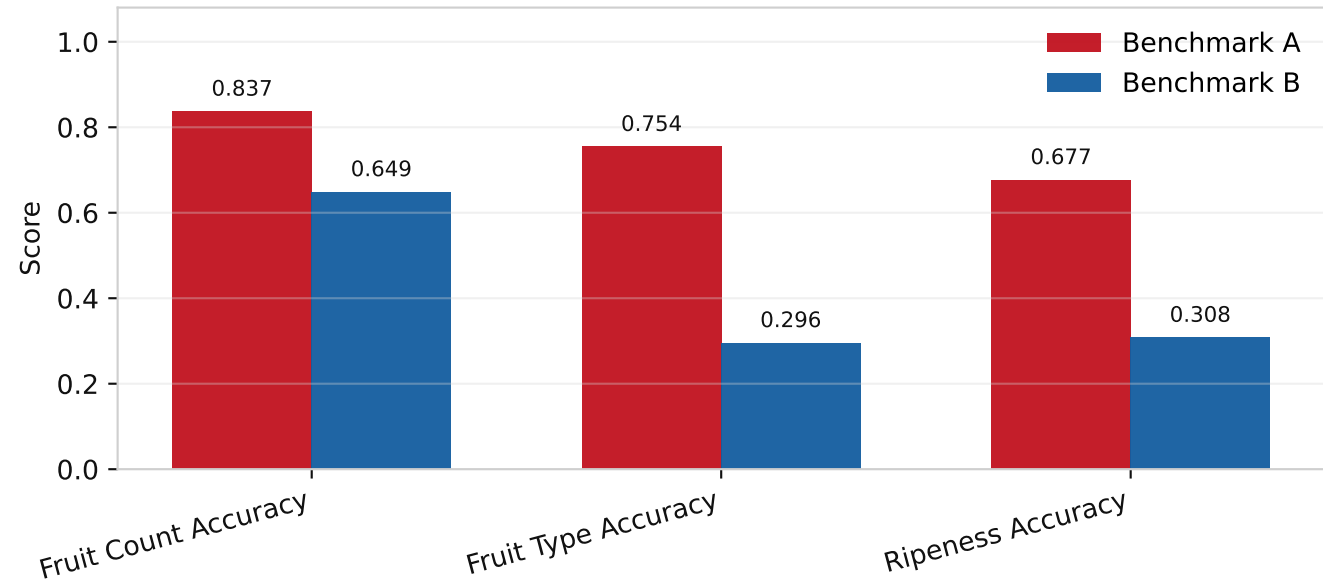
Stage	Benchmark A	Benchmark B
Preprocessing	Median Filter 3×3	Wiener Filter 3×3
Segmentation	Otsu Thresholding	Global Thresholding, threshold 128
Morphological Processing	Morphological Opening, ellipse 3×3	Morphological Closing, ellipse 3×3
Fruit Detection	External Contours, minimum area 10,000	Connected Component Analysis, 8-connectivity, minimum area 7,000
Fruit Counting	Valid Contour Counting	Connected Component Counting
Feature Extraction	RGB Colour Histogram, 8 bins/channel	Uniform LBP, radius 3, 8 points
Fruit Type Classification	LightGBM	KNN
Ripeness Classification	CatBoost	Decision Tree

Overall Prediction Performance

This table compares the final fruit count, fruit type and ripeness prediction accuracy for Benchmark A and Benchmark B.

Benchmark	Fruit Count Accuracy	Fruit Type Accuracy	Ripeness Accuracy
Benchmark A	0.8370	0.7544	0.6769
Benchmark B	0.6487	0.2955	0.3080

Overall Final Prediction Accuracy

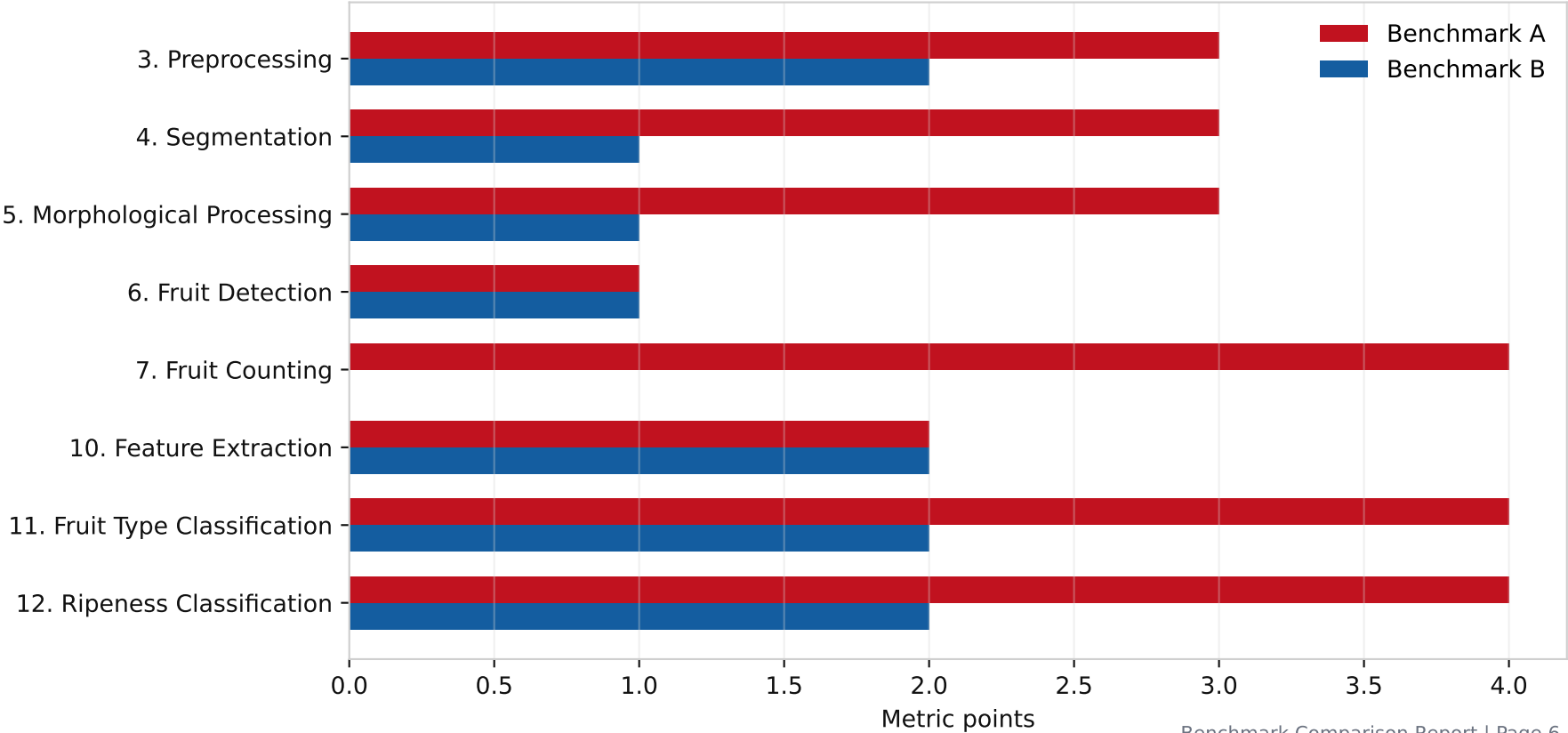


Overall Stage Scorecard

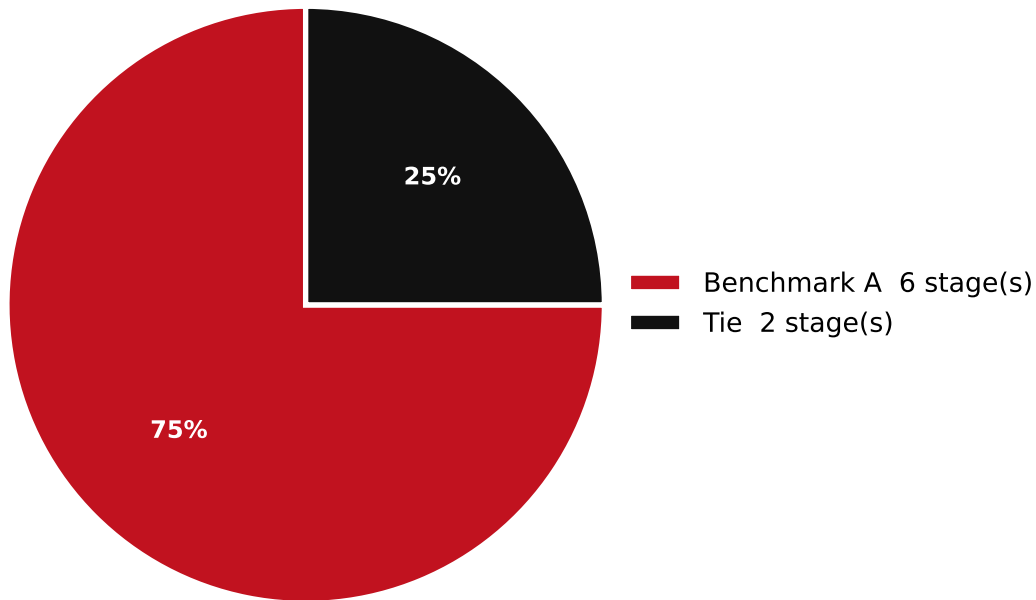
Each available metric gives one comparison point to the better benchmark. Higher is better for quality and accuracy metrics. Lower is better for error and processing time metrics.

Stage	Benchmark A Score	Benchmark B Score	Metrics Compared	Winner
3. Preprocessing	3.0	2.0	5	Benchmark A
4. Segmentation	3.0	1.0	4	Benchmark A
5. Morphological Processing	3.0	1.0	4	Benchmark A
6. Fruit Detection	1.0	1.0	2	Tie
7. Fruit Counting	4.0	0.0	4	Benchmark A
10. Feature Extraction	2.0	2.0	4	Tie
11. Fruit Type Classification	4.0	2.0	6	Benchmark A
12. Ripeness Classification	4.0	2.0	6	Benchmark A

Per Stage Comparison Score



Stage Winners



3. Preprocessing

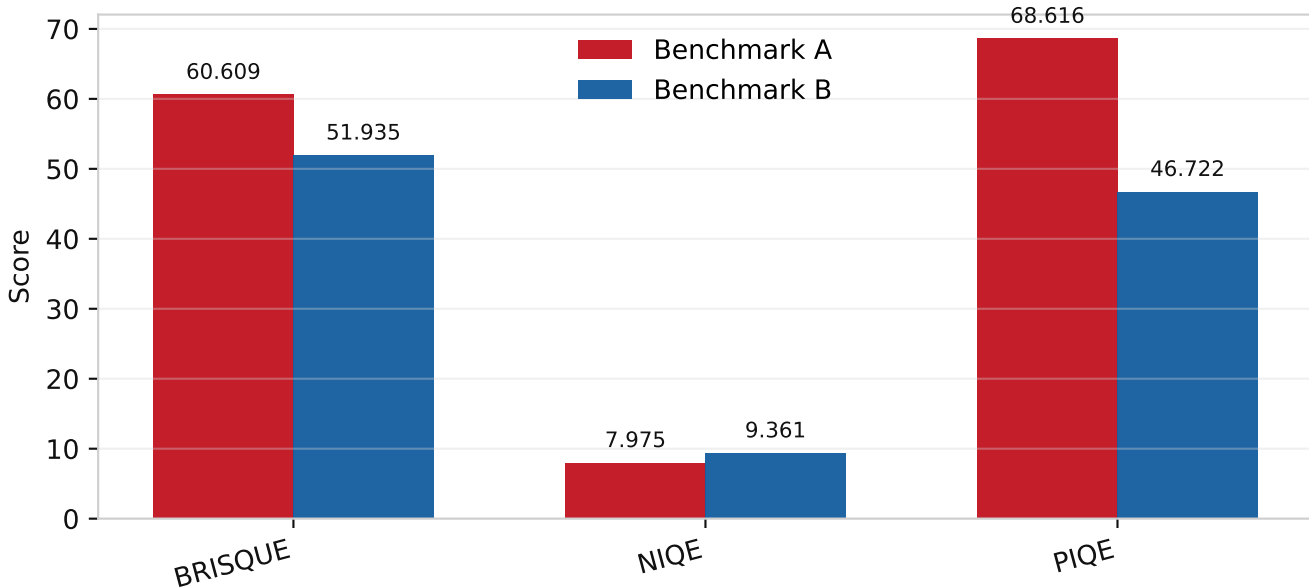
Preprocessing improves the input image before segmentation. Benchmark A uses a Median Filter 3×3, while Benchmark B uses a Wiener Filter 3×3. BRISQUE, NIQE and PIQE are better when lower, SSIM is better when higher, and processing time is better when lower.

Stage Winner Benchmark A	Benchmark A Score 3.0	Benchmark B Score 2.0	Best SSIM 0.9637
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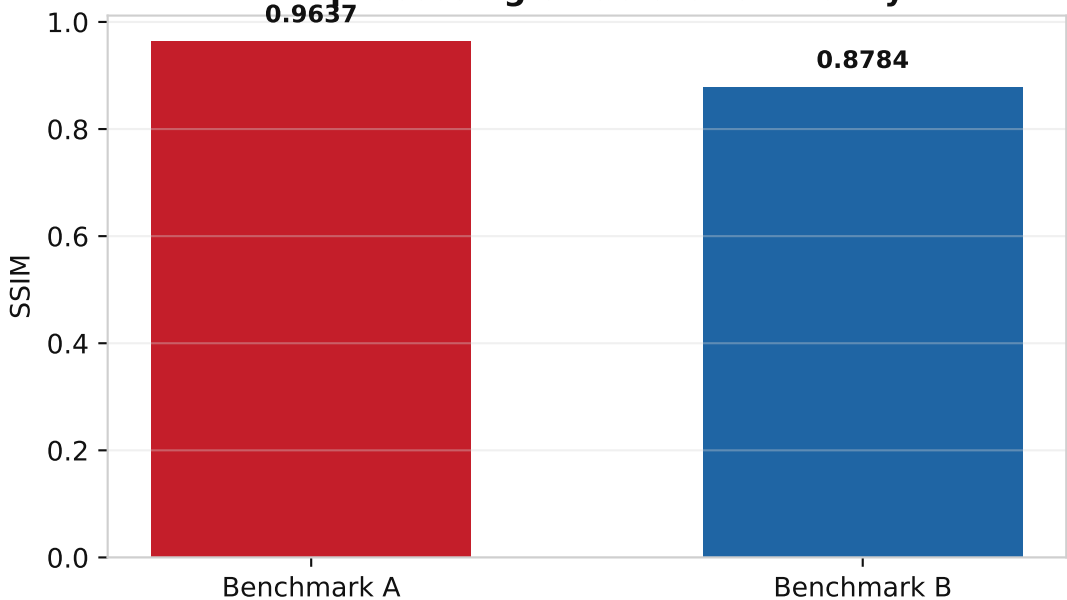
Benchmark A is the overall winner for this stage based on 5 available performance metrics. The strongest primary result is SSIM, achieved by Benchmark A at 0.9637. Higher is better.

Benchmark	Technique	BRISQUE	NIQE	PIQE	SSIM	Time (ms)
Benchmark A	Median Filter 3×3	60.6086	7.9747	68.6163	0.9637	0.1807
Benchmark B	Wiener Filter 3×3	51.9348	9.3613	46.7222	0.8784	164.9897

Preprocessing No Reference Image Quality

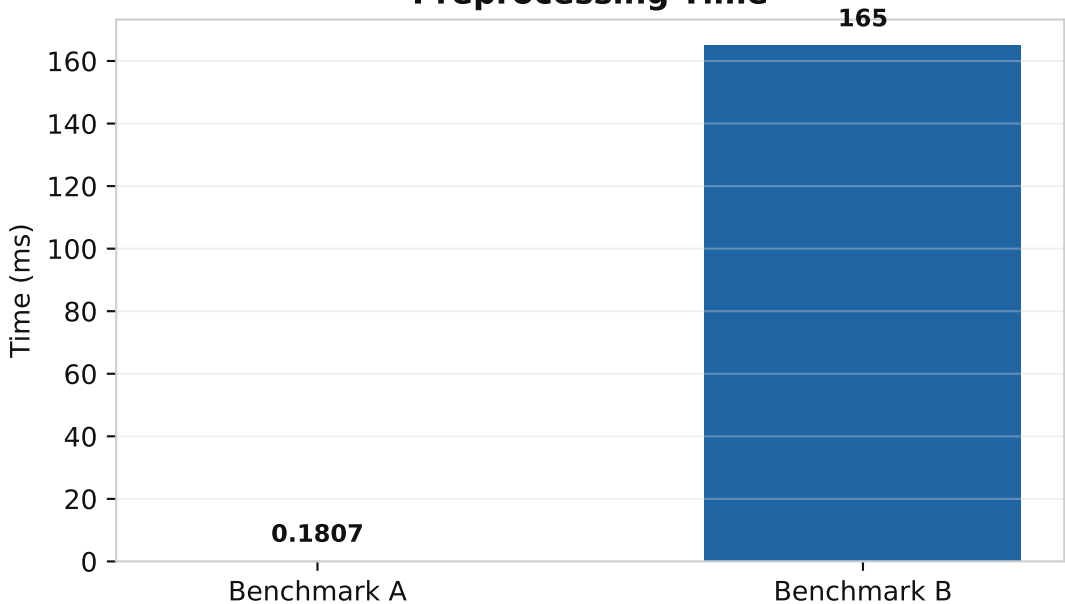


Preprocessing Structural Similarity



Higher is better

Preprocessing Time



Lower is better

4. Segmentation

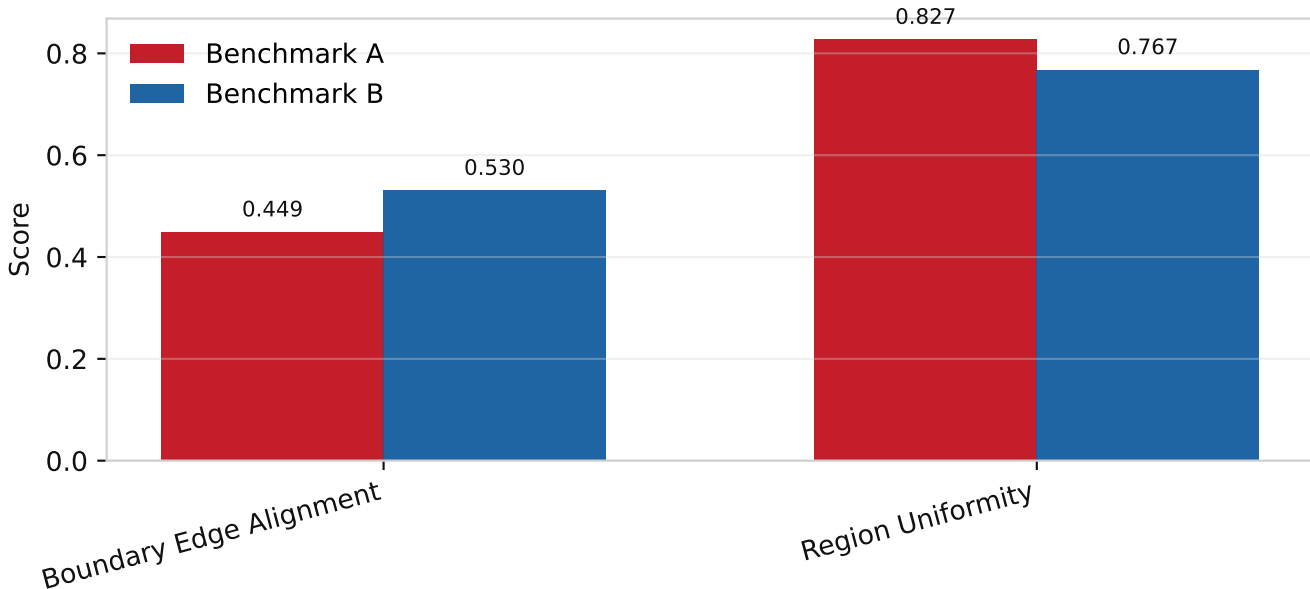
Segmentation separates the fruit region from the background. Benchmark A uses Otsu Thresholding, while Benchmark B uses Global Thresholding with threshold 128. The comparison focuses on boundary edge alignment, region uniformity, foreground background contrast and processing time.

Stage Winner Benchmark A	Benchmark A Score 3.0	Benchmark B Score 1.0	Best Boundary Edge Alignment 0.5303
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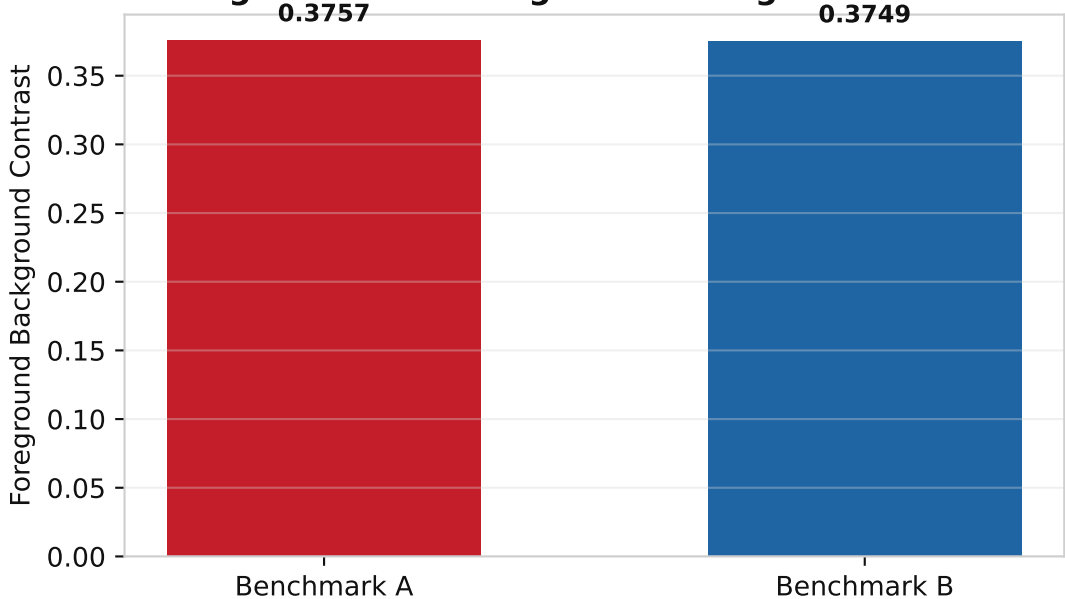
Benchmark A is the overall winner for this stage based on 4 available performance metrics. The strongest primary result is Boundary Edge Alignment, achieved by Benchmark B at 0.5303. Higher is better.

Benchmark	Technique	Boundary Edge Alignment	Region Uniformity	Foreground Background Contrast	Time (ms)
Benchmark A	Otsu Thresholding	0.4487	0.8269	0.3757	0.2881
Benchmark B	Global Thresholding, threshold 128	0.5303	0.7674	0.3749	0.6782

4. Segmentation Mask Quality

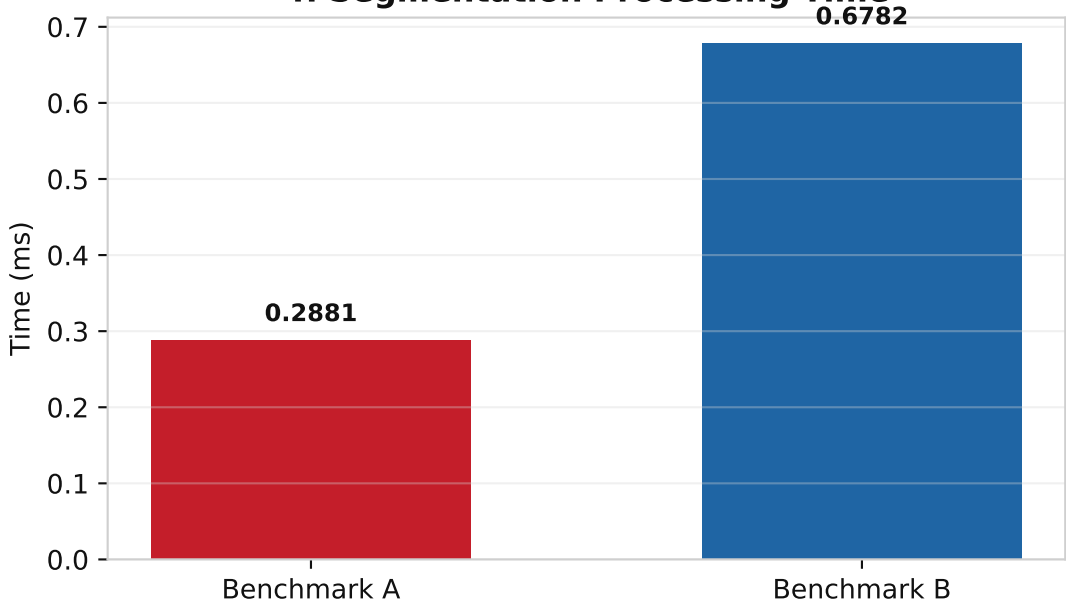


4. Segmentation Foreground Background Contrast



Higher is better

4. Segmentation Processing Time



Lower is better

5. Morphological Processing

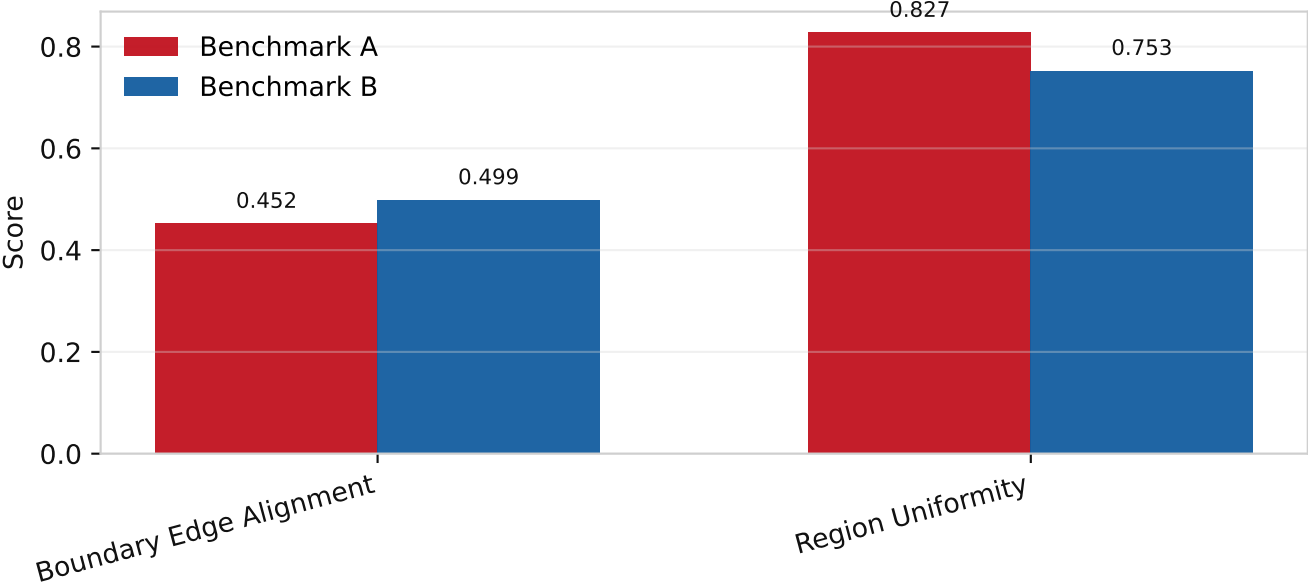
Morphological processing refines the segmented mask. Benchmark A uses Morphological Opening with an ellipse 3×3 kernel, while Benchmark B uses Morphological Closing with an ellipse 3×3 kernel. Higher mask quality scores are preferred and lower processing time is preferred.

Stage Winner Benchmark A	Benchmark A Score 3.0	Benchmark B Score 1.0	Best Boundary Edge Alignment 0.4985
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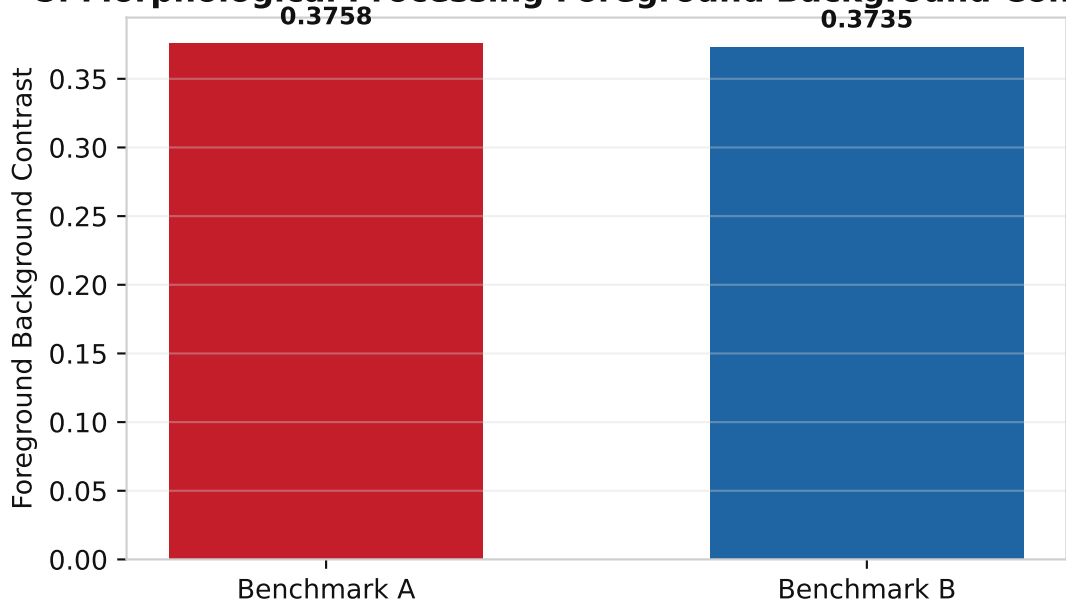
Benchmark A is the overall winner for this stage based on 4 available performance metrics. The strongest primary result is Boundary Edge Alignment, achieved by Benchmark B at 0.4985. Higher is better.

Benchmark	Technique	Boundary Edge Alignment	Region Uniformity	Foreground Background Contrast	Time (ms)
Benchmark A	Morphological Opening, ellipse 3×3	0.4521	0.8274	0.3758	0.0750
Benchmark B	Morphological Closing, ellipse 3×3	0.4985	0.7526	0.3735	0.3518

5. Morphological Processing Mask Quality

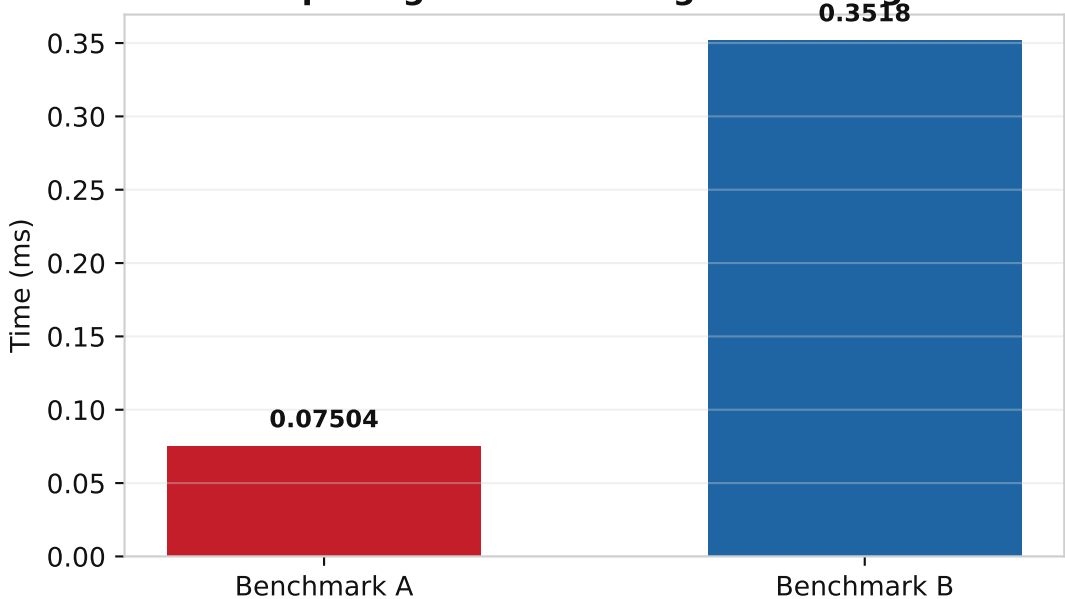


5. Morphological Processing Foreground Background Contrast



Higher is better

5. Morphological Processing Processing Time



Lower is better

6. Fruit Detection

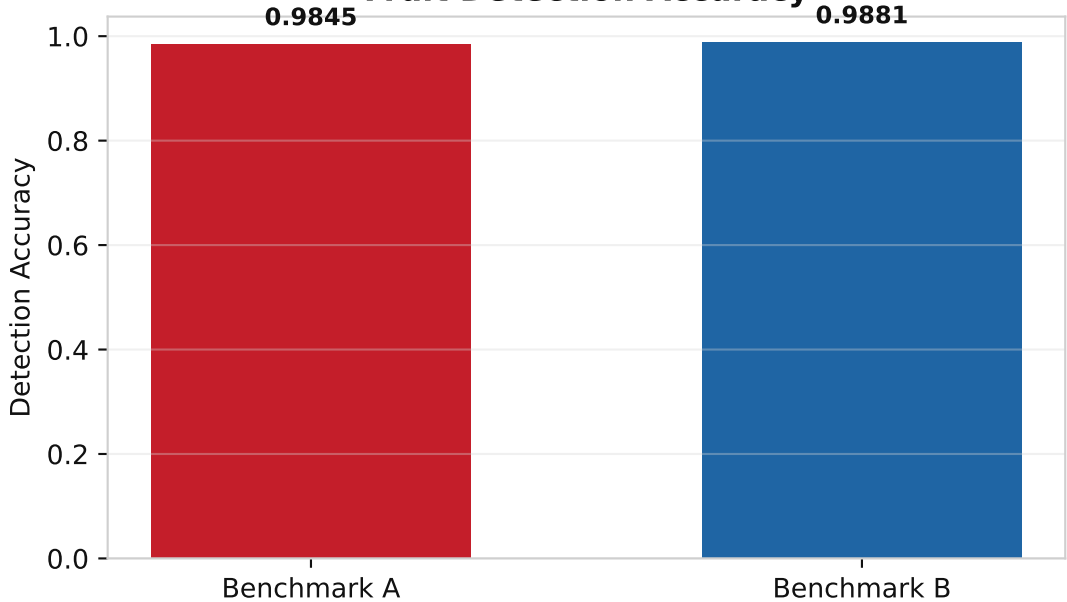
Fruit detection identifies valid fruit objects after mask refinement. Benchmark A uses External Contour Detection with minimum area 10,000. Benchmark B uses Connected Component Analysis with 8-connectivity and minimum area 7,000. Higher detection accuracy and lower processing time are preferred.

Stage Winner Tie	Benchmark A Score 1.0	Benchmark B Score 1.0	Best Detection Accuracy 0.9881
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Tie is the overall winner for this stage based on 2 available performance metrics. The strongest primary result is Detection Accuracy, achieved by Benchmark B at 0.9881. Higher is better.

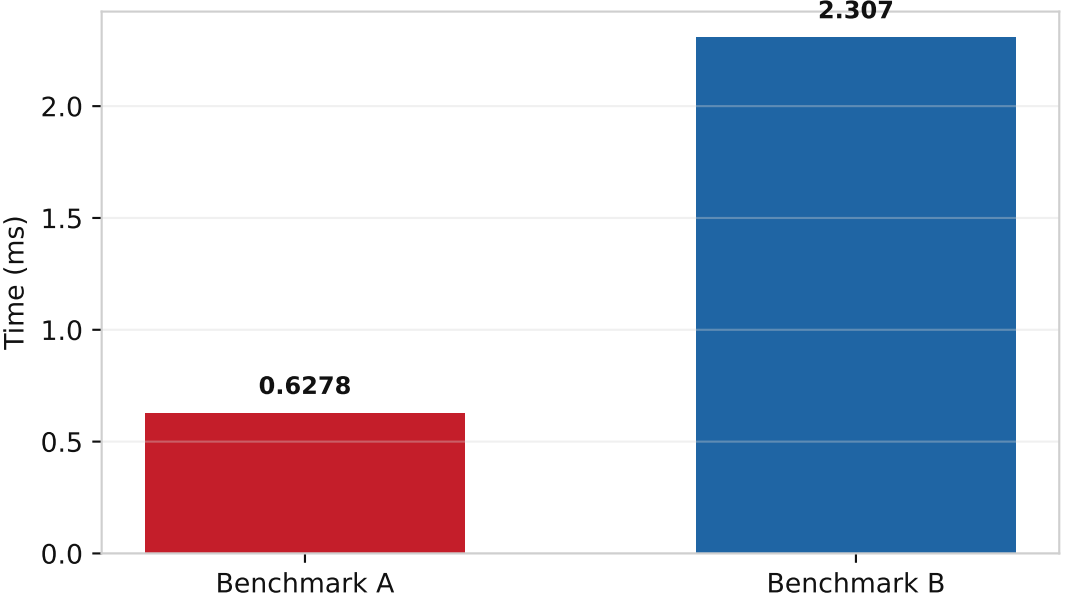
Benchmark	Technique	Detection Accuracy	Time (ms)
Benchmark A	External Contours, minimum area 10,000	0.9845	0.6278
Benchmark B	Connected Component Analysis, 8-connectivity, minimum area 7,000	0.9881	2.3074

Fruit Detection Accuracy



Higher is better

Fruit Detection Time



Lower is better

7. Fruit Counting

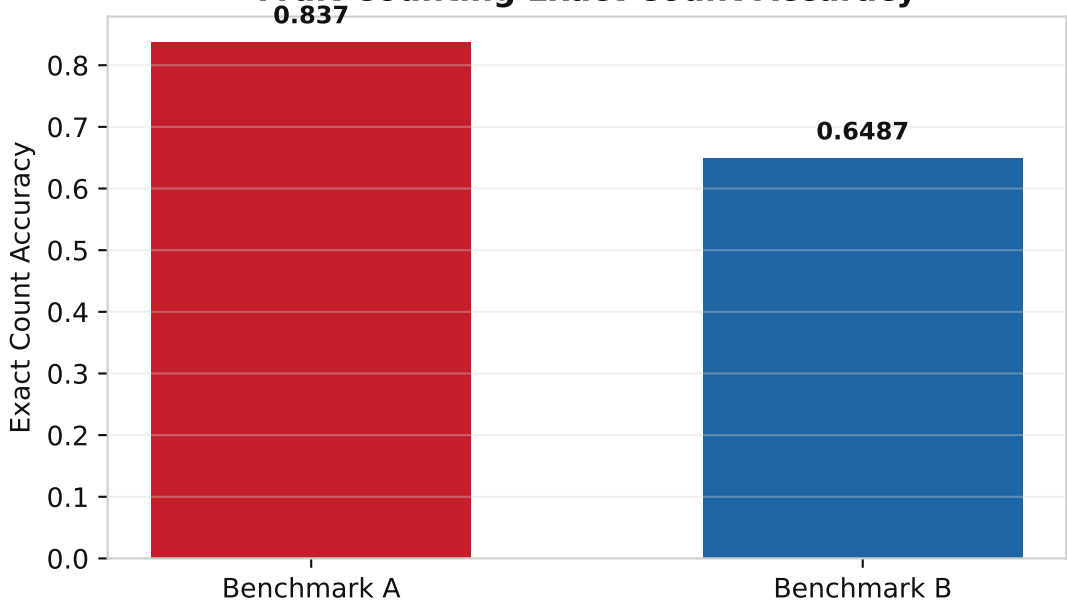
Fruit counting estimates the number of fruit objects in each image. Benchmark A uses Valid Contour Counting, while Benchmark B uses Connected Component Counting. Higher Exact Count Accuracy is preferred, while lower MAE, MAPE and processing time are preferred.

Stage Winner Benchmark A	Benchmark A Score 4.0	Benchmark B Score 0.0	Best Exact Count Accuracy 0.8370
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Benchmark A is the overall winner for this stage based on 4 available performance metrics. The strongest primary result is Exact Count Accuracy, achieved by Benchmark A at 0.8370. Higher is better.

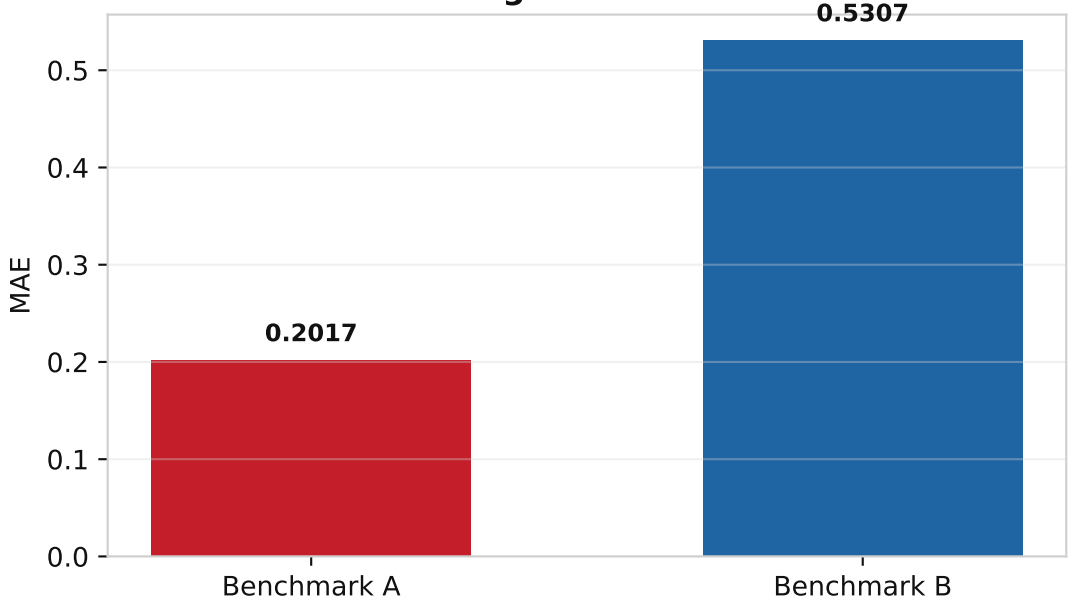
Benchmark	Technique	Exact Count Accuracy	MAE	MAPE	Time (ms)
Benchmark A	Valid Contour Counting	0.8370	0.2017	0.2017	0.0001
Benchmark B	Connected Component Counting	0.6487	0.5307	0.5307	0.0001

Fruit Counting Exact Count Accuracy



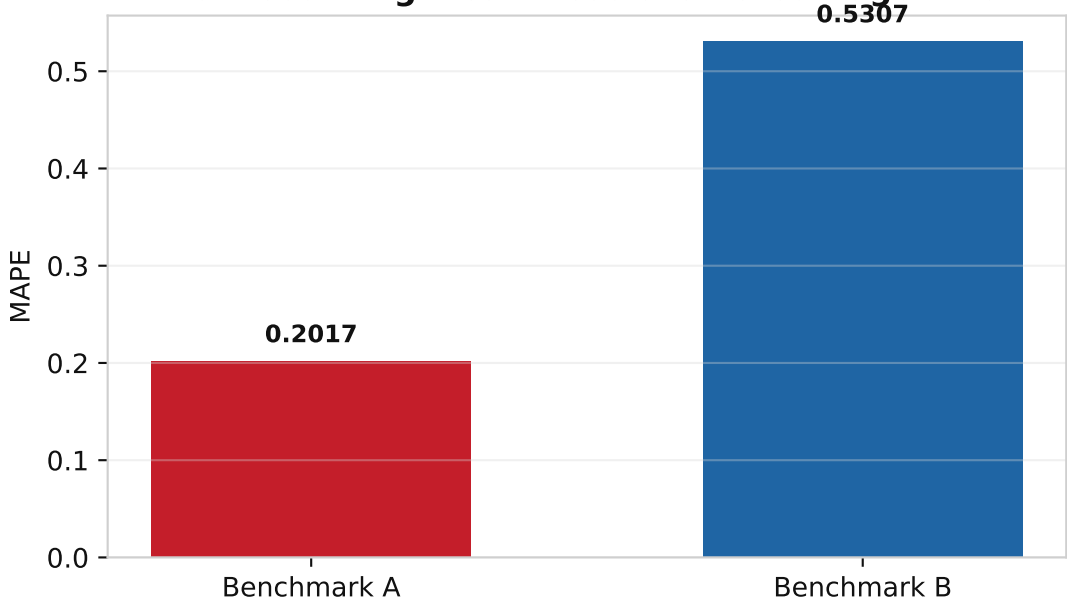
Higher is better

Fruit Counting Mean Absolute Error



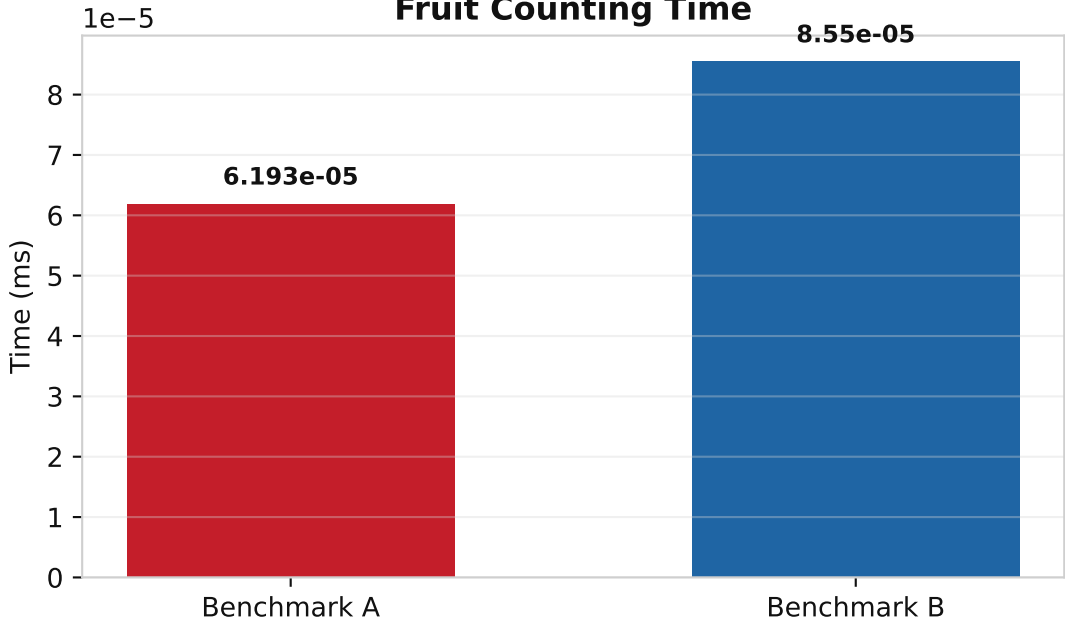
Lower is better

Fruit Counting Mean Absolute Percentage Error



Lower is better

Fruit Counting Time



Lower is better

10. Feature Extraction

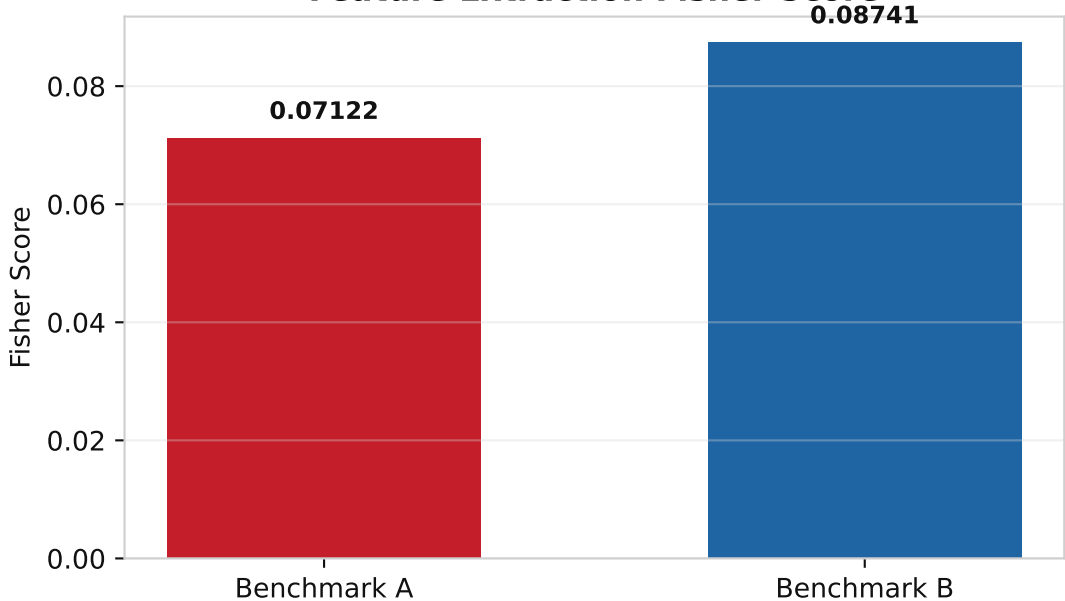
Feature extraction converts each fruit ROI into a numerical feature vector. Benchmark A uses an RGB Colour Histogram with 8 bins per channel. Benchmark B uses Uniform LBP with radius 3 and 8 points. Higher Fisher and Silhouette scores indicate stronger separation, while lower extraction time and smaller feature size improve efficiency.

Stage Winner Tie	Benchmark A Score 2.0	Benchmark B Score 2.0	Best Fisher Score 0.0874
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Tie is the overall winner for this stage based on 4 available performance metrics. The strongest primary result is Fisher Score, achieved by Benchmark B at 0.0874. Higher is better.

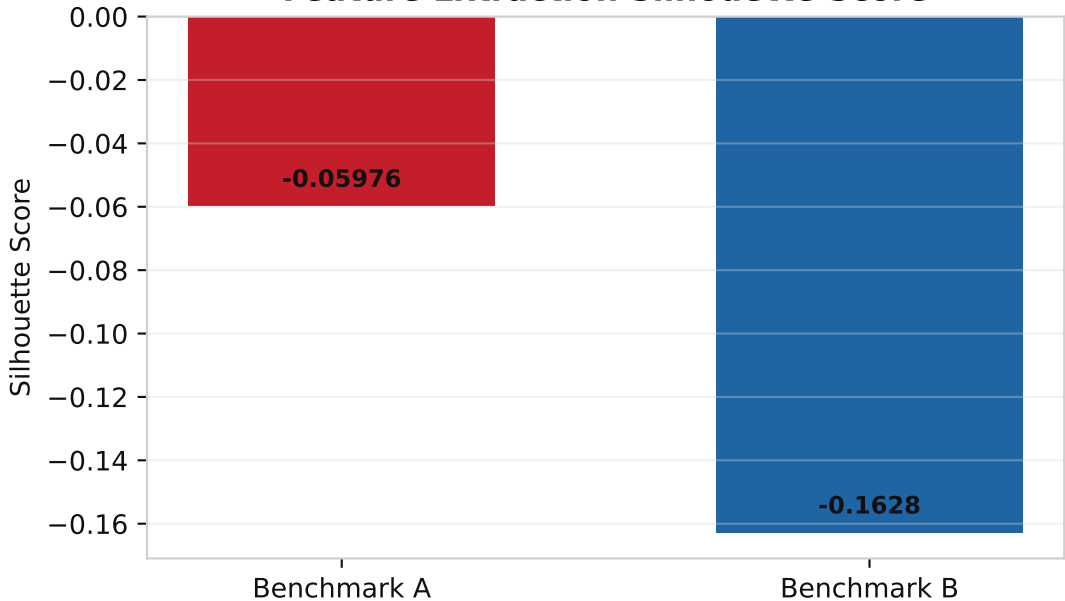
Benchmark	Technique	Fisher Score	Silhouette Score	Time (ms/ROI)	Feature Size
Benchmark A	RGB Colour Histogram, 8 bins/channel	0.0712	-0.0598	2.9490	24
Benchmark B	Uniform LBP, radius 3, 8 points	0.0874	-0.1628	51.0775	10

Feature Extraction Fisher Score



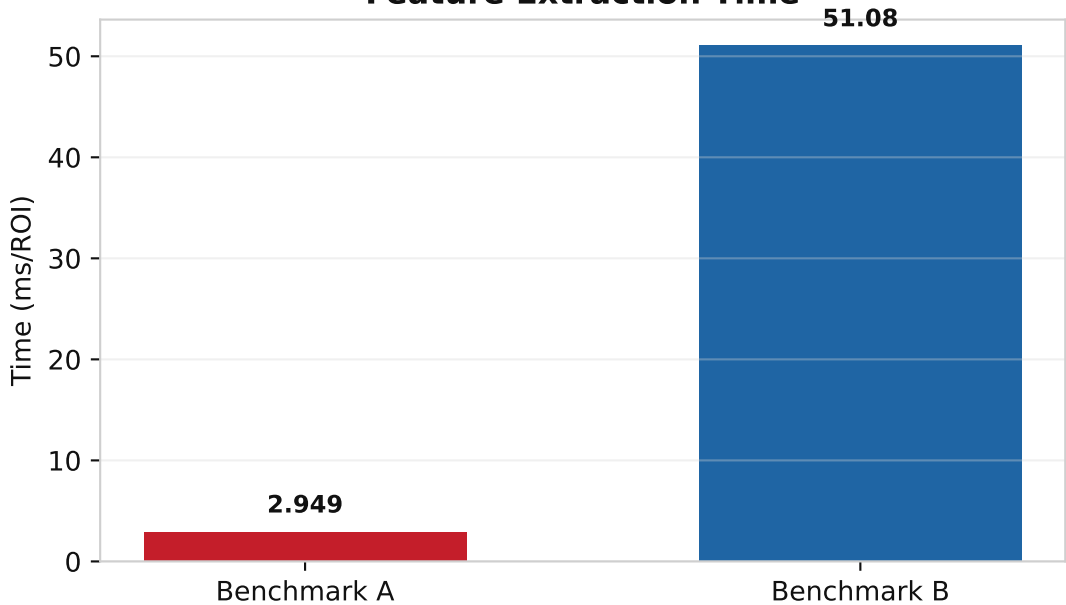
Higher is better

Feature Extraction Silhouette Score



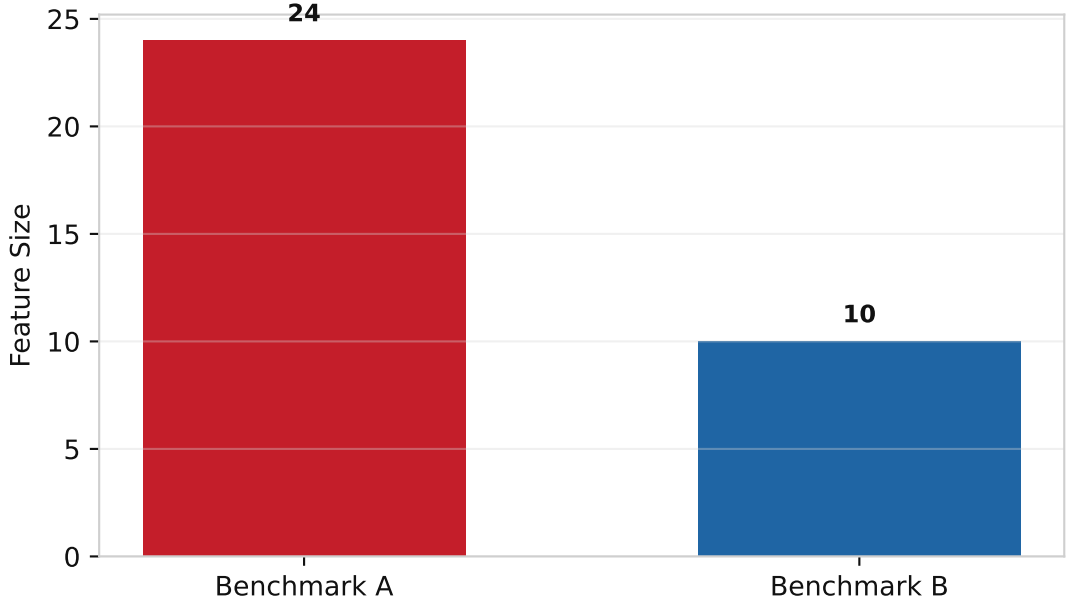
Higher is better

Feature Extraction Time



Lower is better

Feature Vector Size



Lower is better

11. Fruit Type Classification

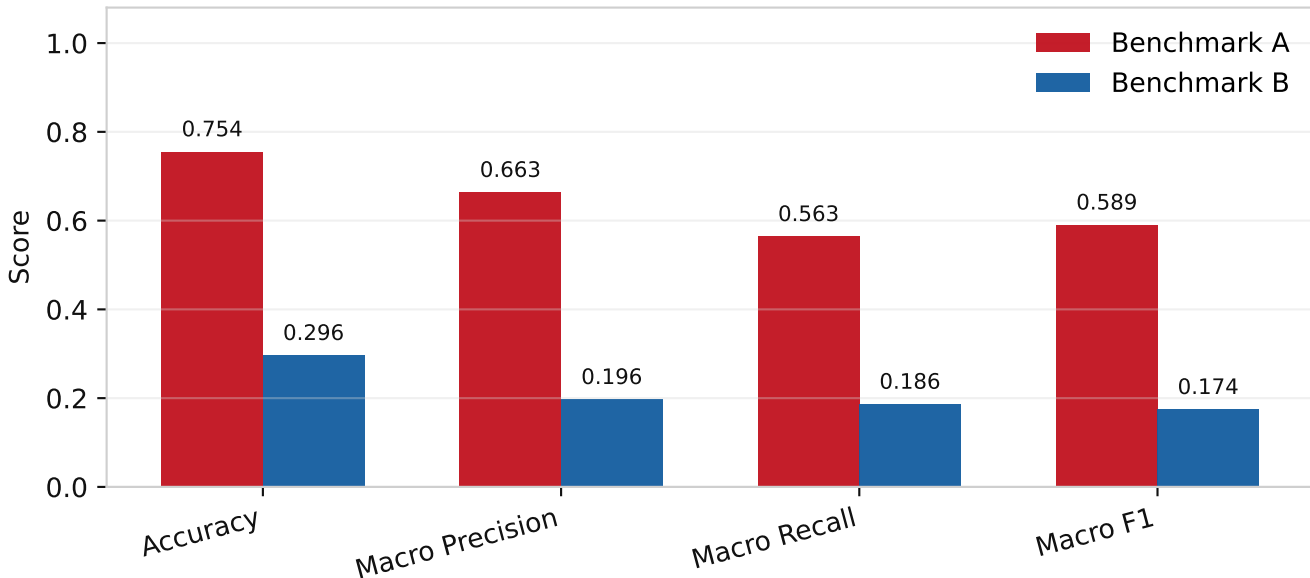
Fruit type classification predicts the fruit category for each detected ROI. Benchmark A uses LightGBM, while Benchmark B uses KNN. Accuracy, Macro Precision, Macro Recall and Macro F1 measure predictive quality, while training and prediction time measure efficiency.

Stage Winner Benchmark A	Benchmark A Score 4.0	Benchmark B Score 2.0	Best Macro F1 0.5894
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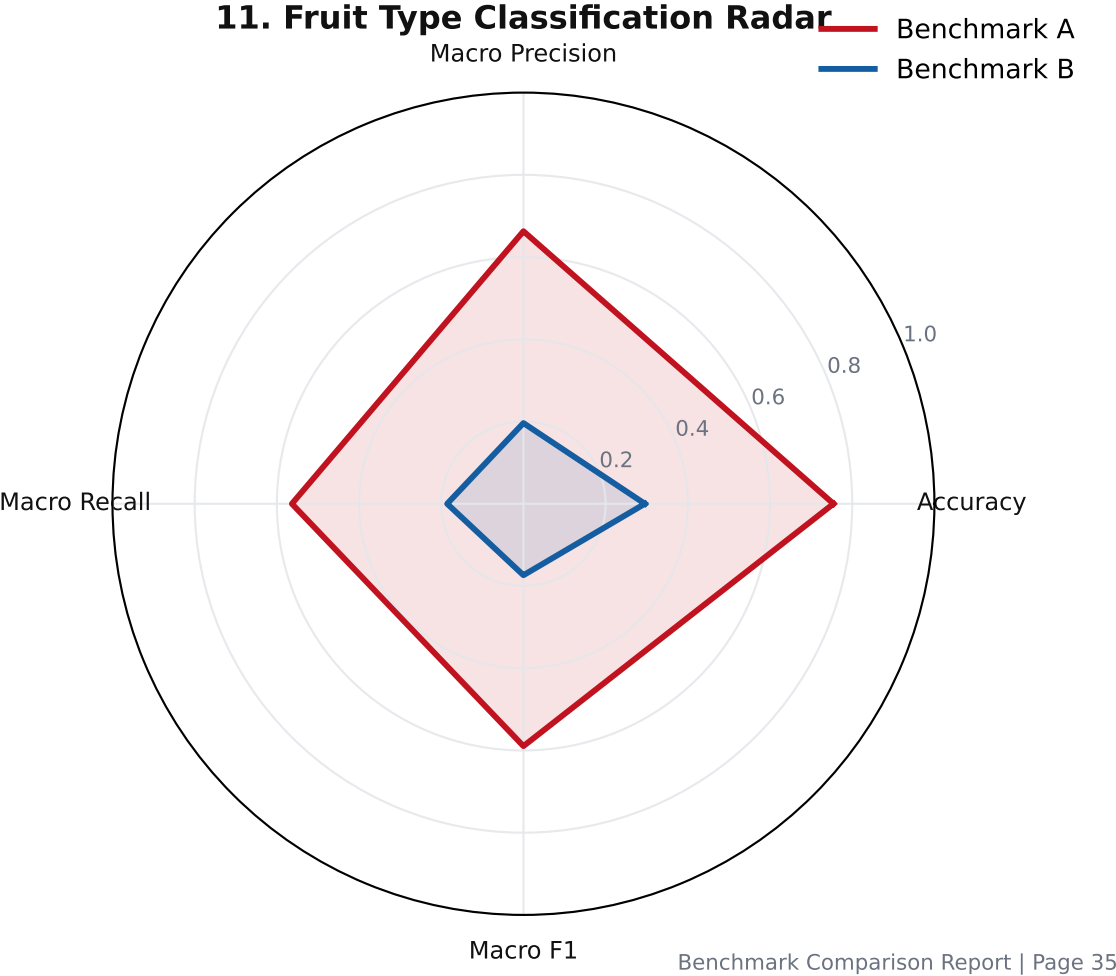
Benchmark A is the overall winner for this stage based on 6 available performance metrics. The strongest primary result is Macro F1, achieved by Benchmark A at 0.5894. Higher is better.

Benchmark	Technique	Accuracy	Macro Precision	Macro Recall	Macro F1	Training Time	Prediction Time (ms/ROI)	Test ROIs
Benchmark A	LightGBM	0.7544	0.6629	0.5634	0.5894	7.1479	0.0364	7,660
Benchmark B	KNN	0.2955	0.1963	0.1856	0.1738	0.1535	0.0201	9,702

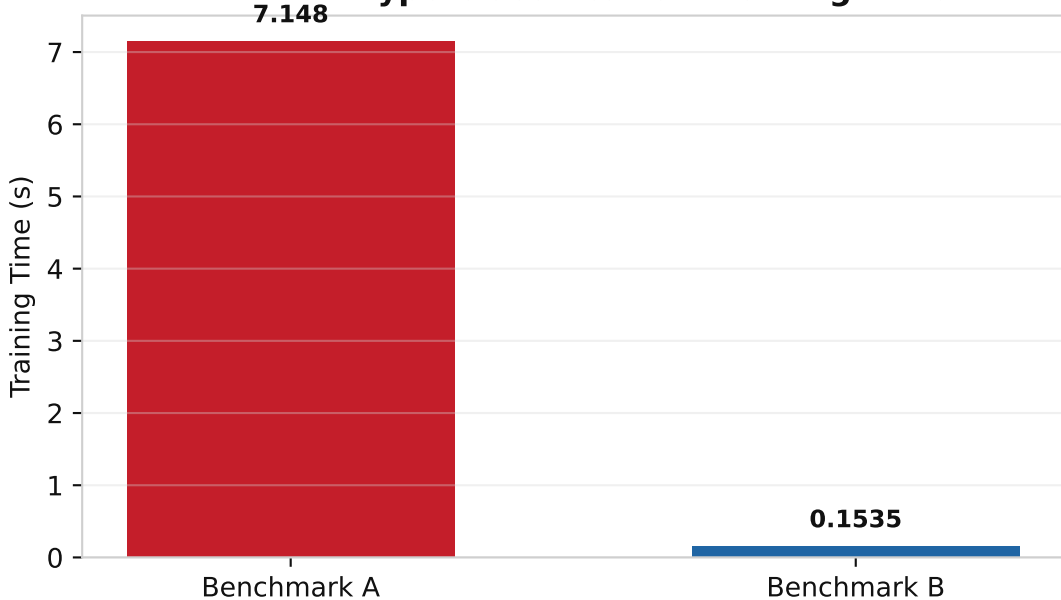
11. Fruit Type Classification Performance



11. Fruit Type Classification Radar

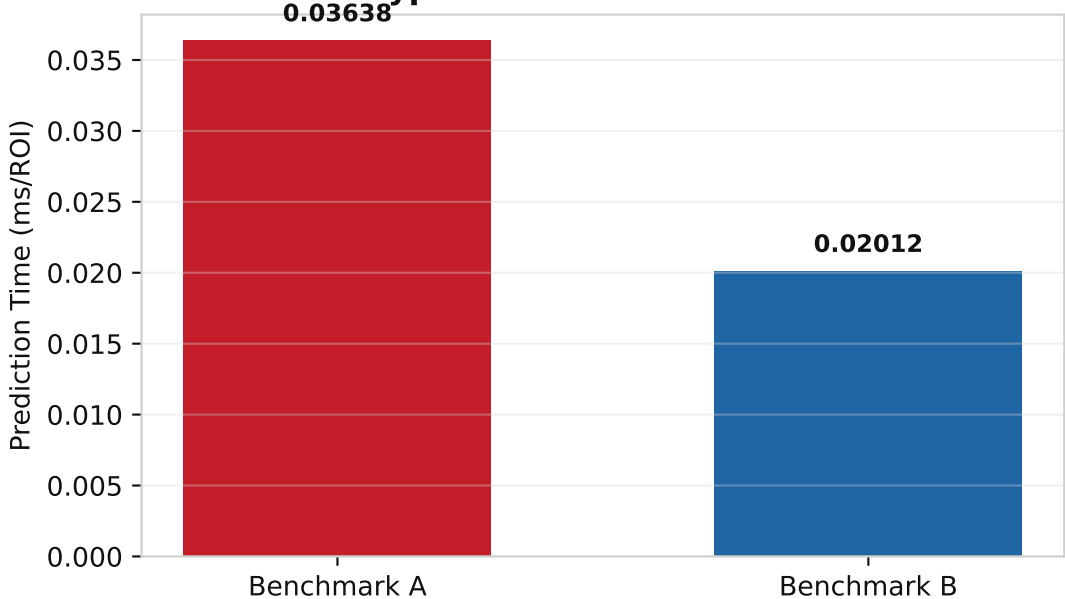


11. Fruit Type Classification Training Time



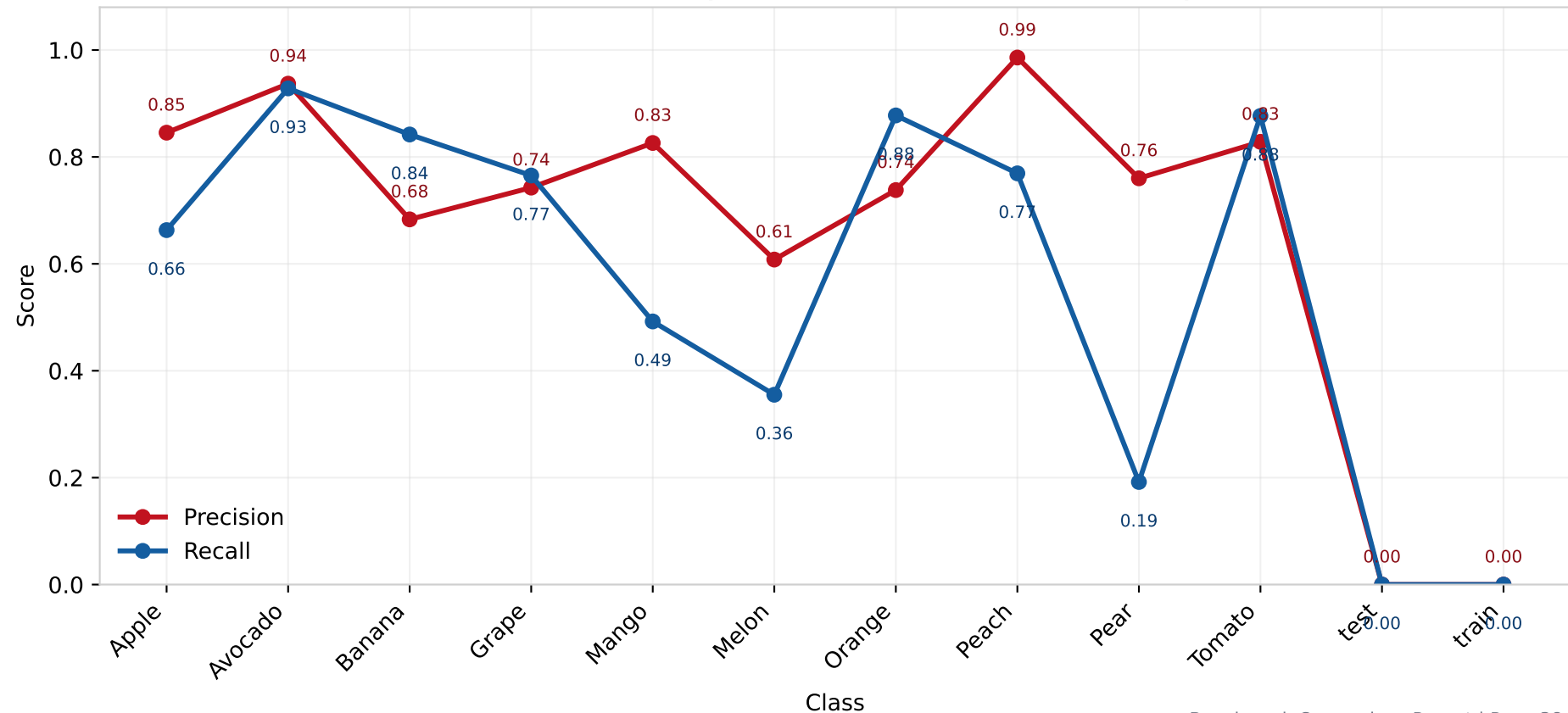
Lower is better

11. Fruit Type Classification Prediction Time

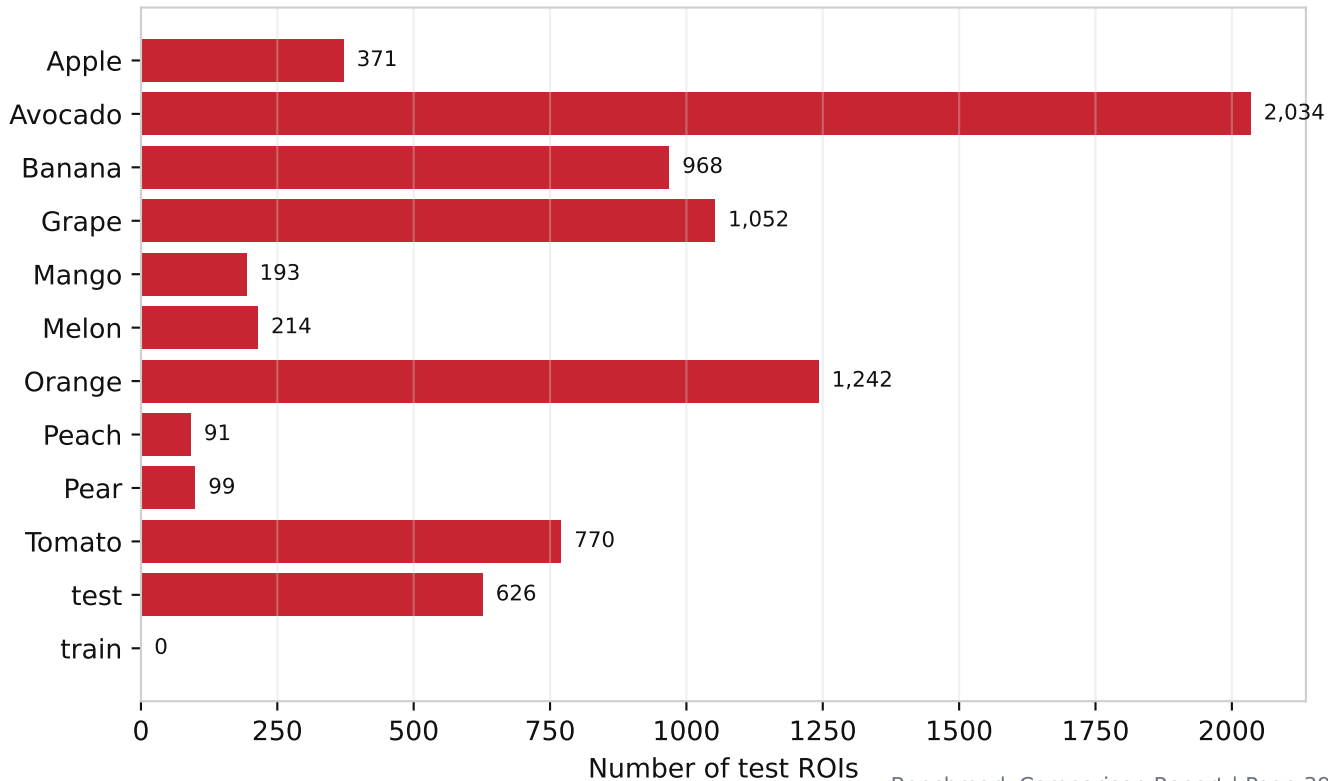


Lower is better

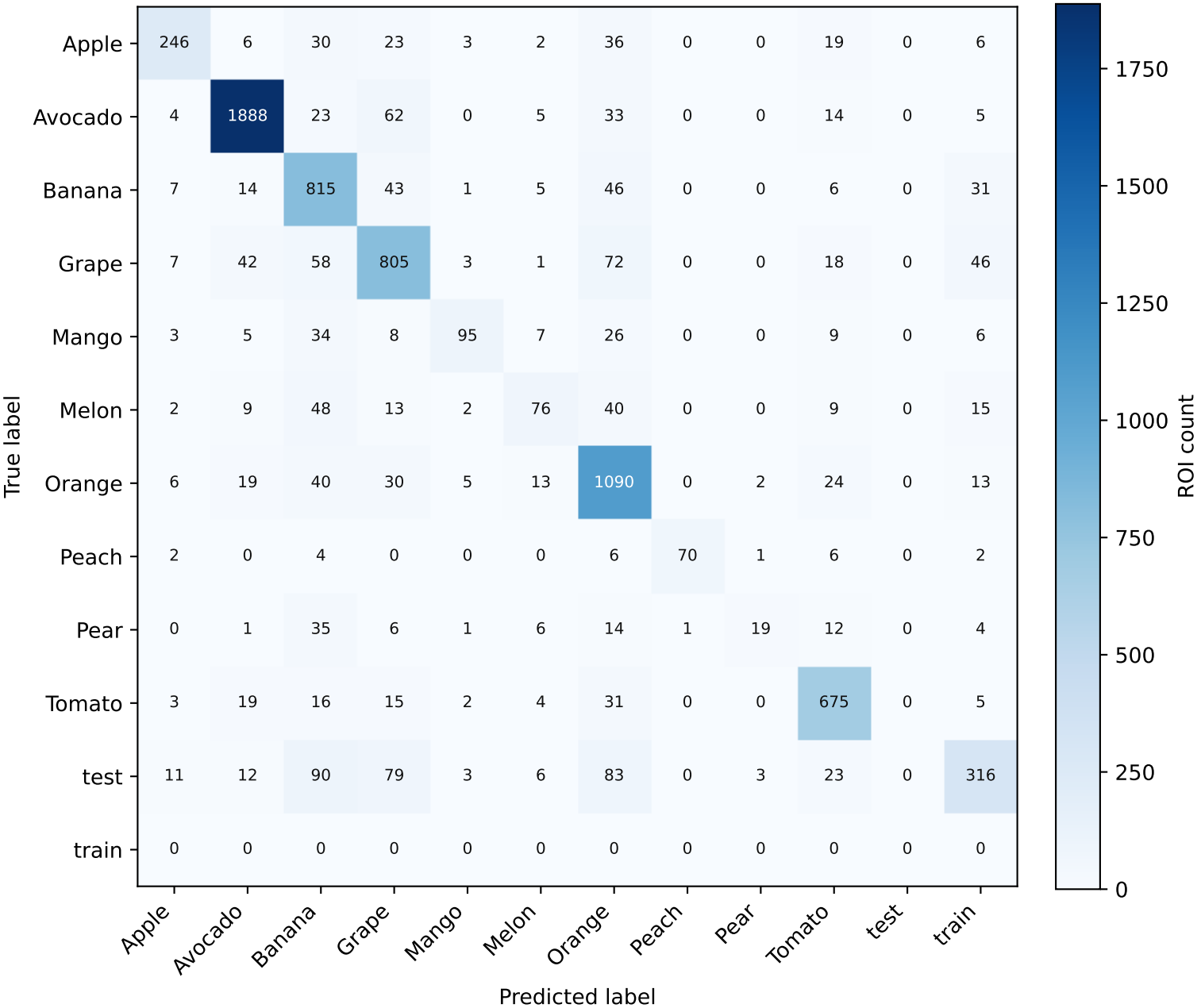
Benchmark A Fruit Type — Precision and Recall by Class



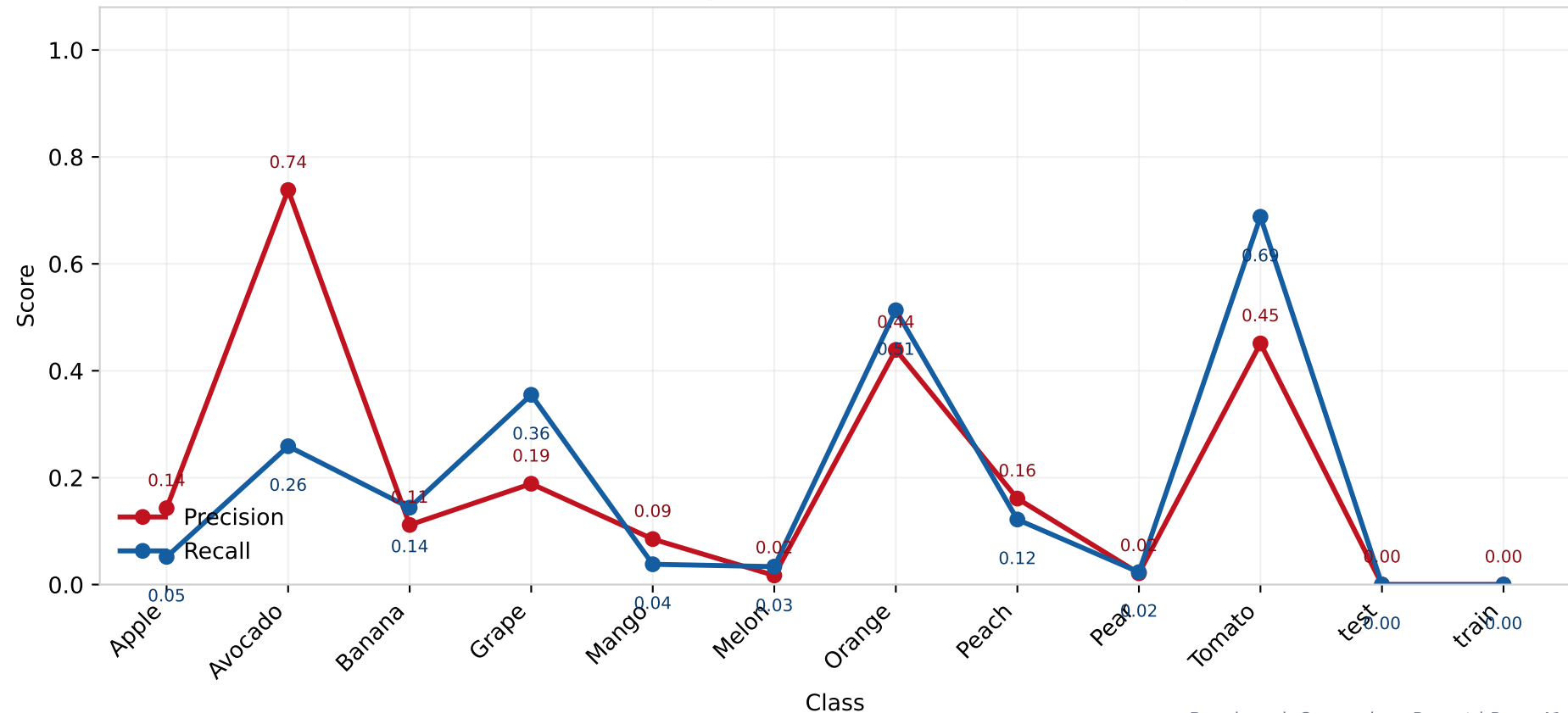
Benchmark A Fruit Type Test Support



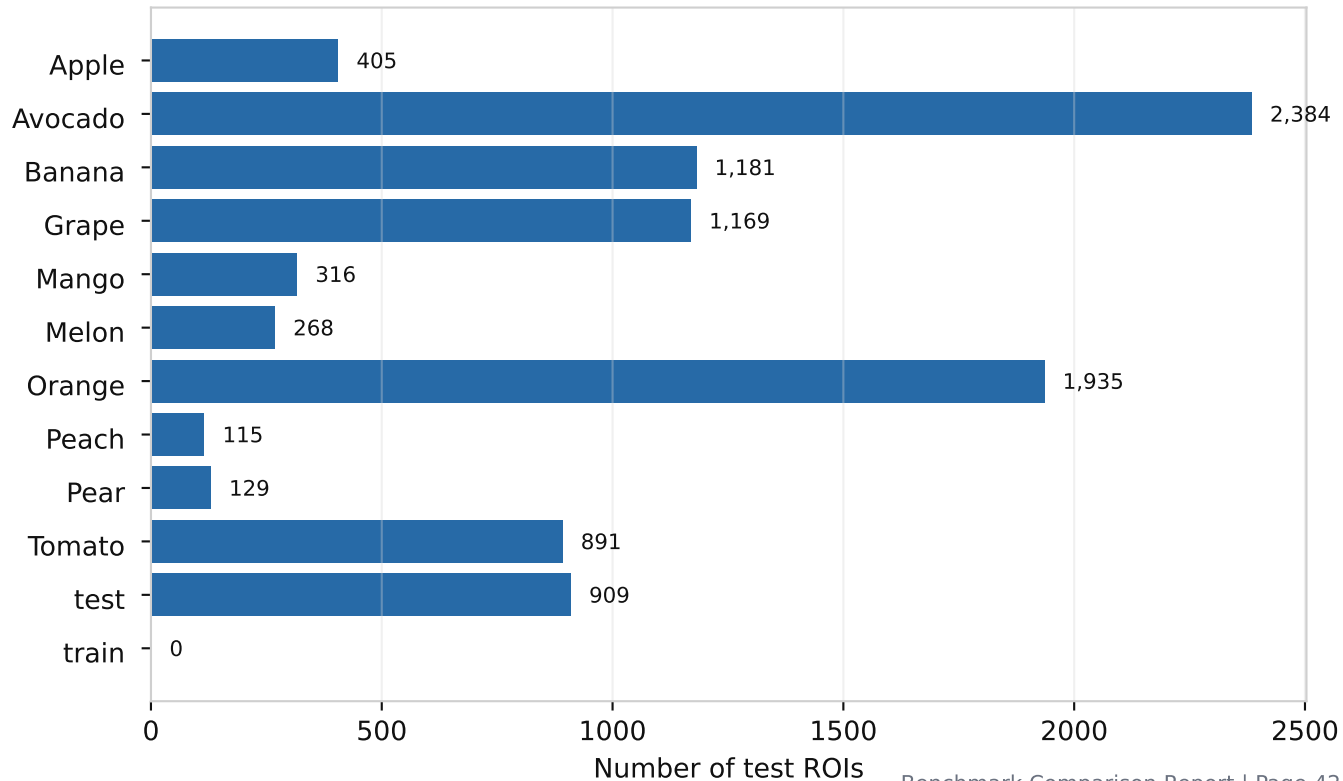
Benchmark A Fruit Type Confusion Matrix



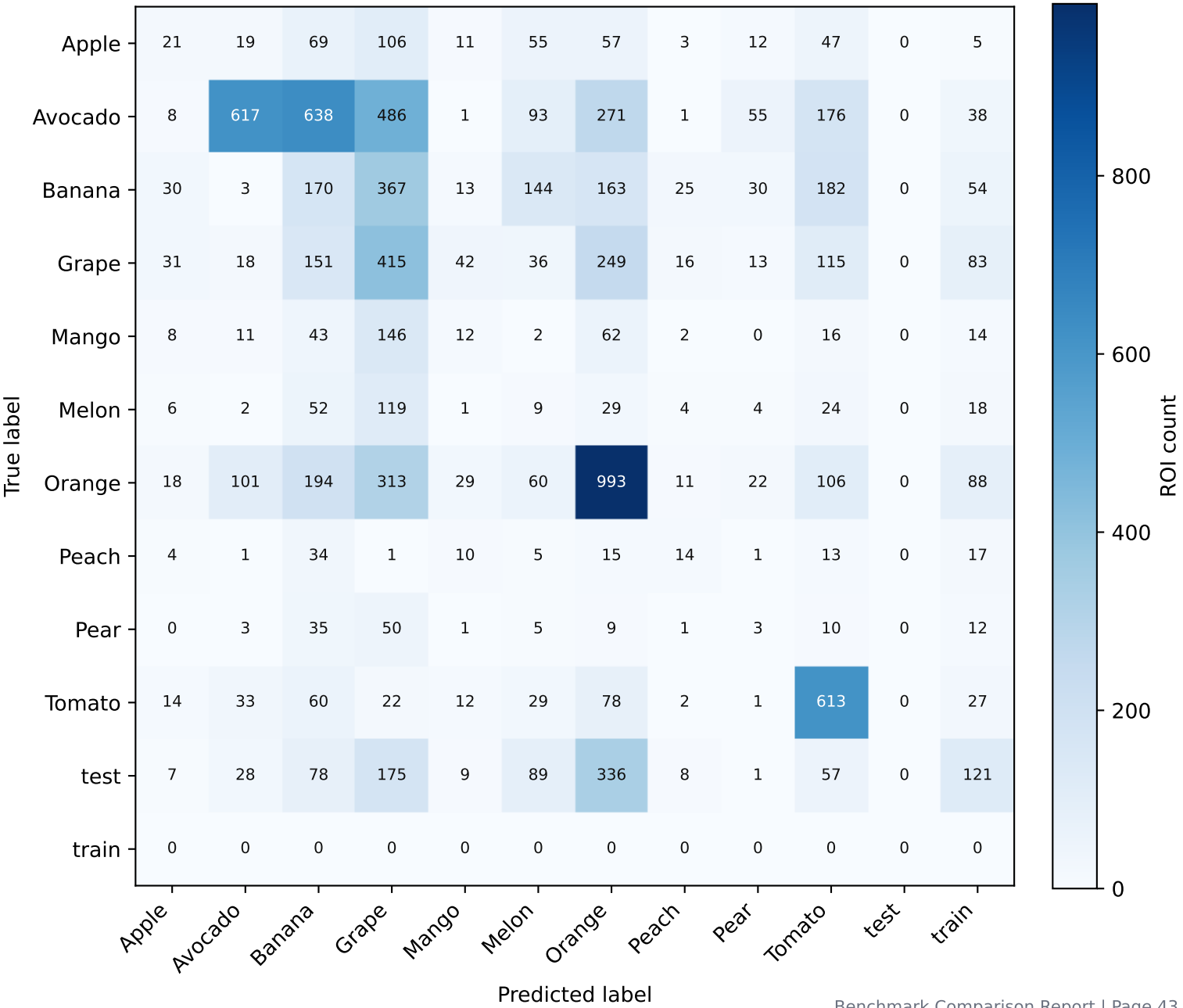
Benchmark B Fruit Type — Precision and Recall by Class



Benchmark B Fruit Type Test Support



Benchmark B Fruit Type Confusion Matrix



12. Ripeness Classification

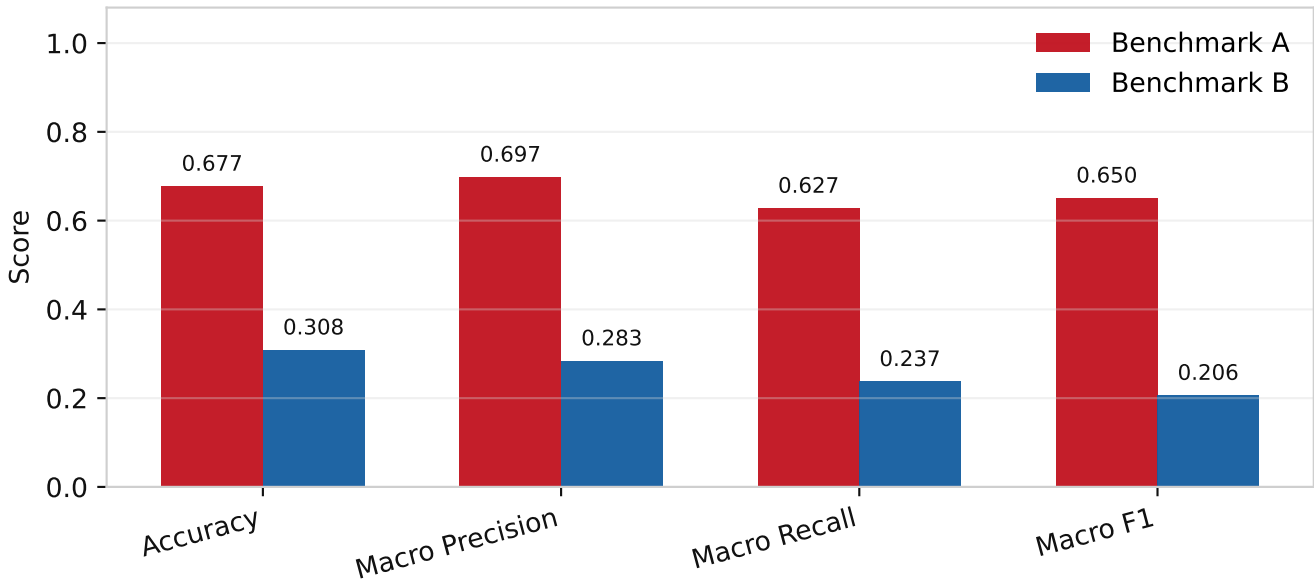
Ripeness classification predicts the ripeness level for each detected fruit ROI. Benchmark A uses CatBoost, while Benchmark B uses a Decision Tree. The models are compared using Accuracy, Macro Precision, Macro Recall, Macro F1, training time, prediction time and the confusion matrix.

Stage Winner Benchmark A	Benchmark A Score 4.0	Benchmark B Score 2.0	Best Macro F1 0.6503
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Benchmark A is the overall winner for this stage based on 6 available performance metrics. The strongest primary result is Macro F1, achieved by Benchmark A at 0.6503. Higher is better.

Benchmark	Technique	Accuracy	Macro Precision	Macro Recall	Macro F1	Training Time	Prediction Time (ms/ROI)	Test ROIs
Benchmark A	CatBoost	0.6769	0.6974	0.6272	0.6503	30.1967	0.0011	7,660
Benchmark B	Decision Tree	0.3080	0.2834	0.2368	0.2057	0.2745	0.0004	9,702

12. Ripeness Classification Performance

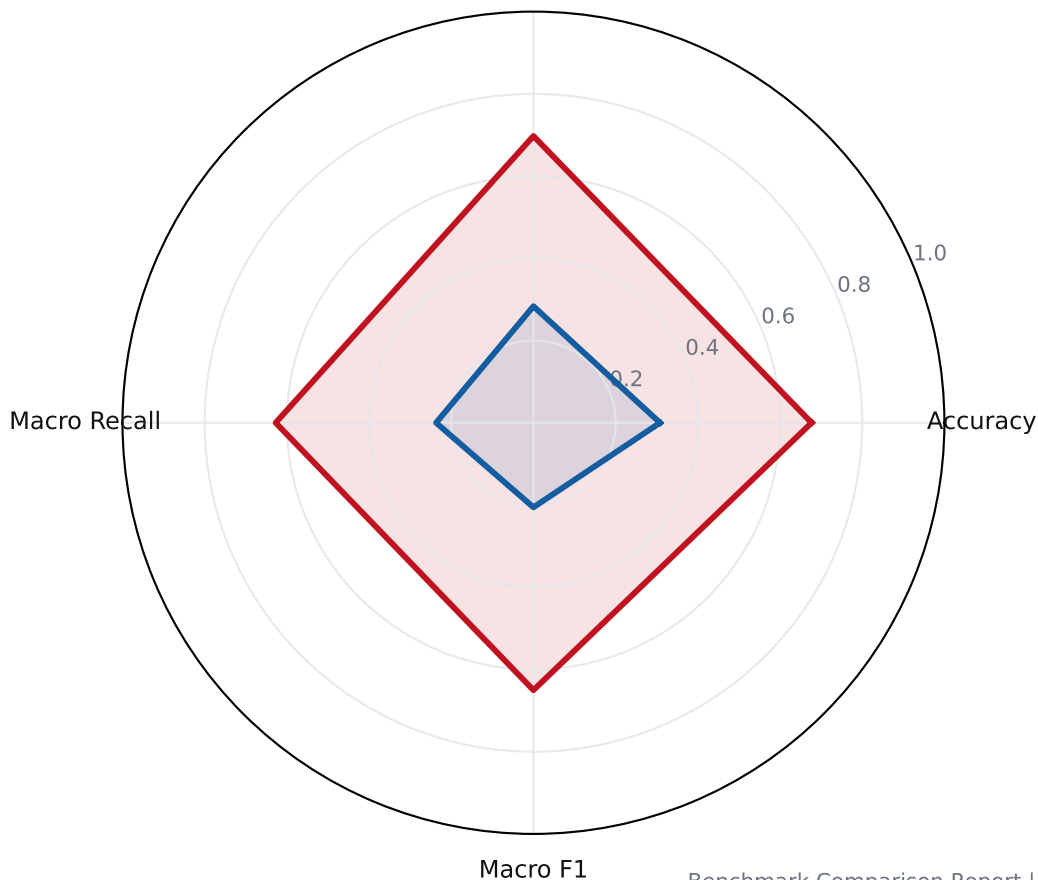


12. Ripeness Classification Radar

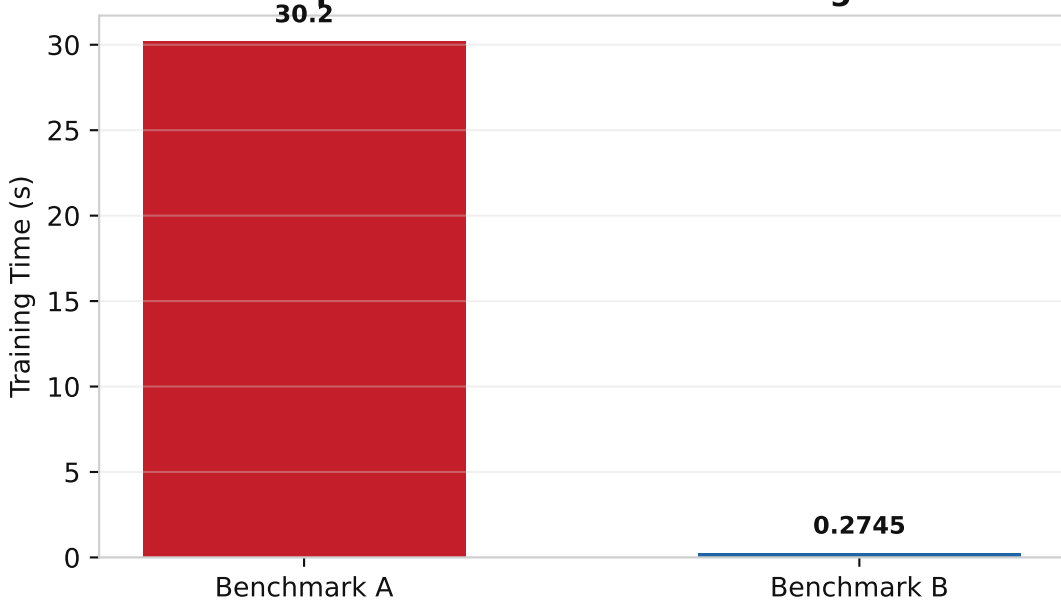
Macro Precision

Benchmark A

Benchmark B

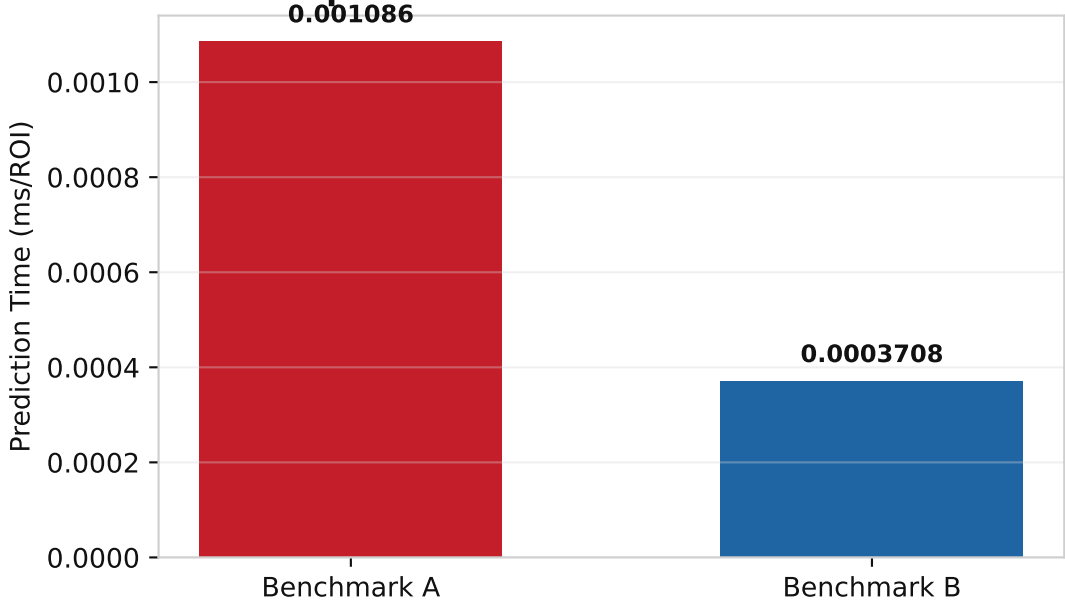


12. Ripeness Classification Training Time



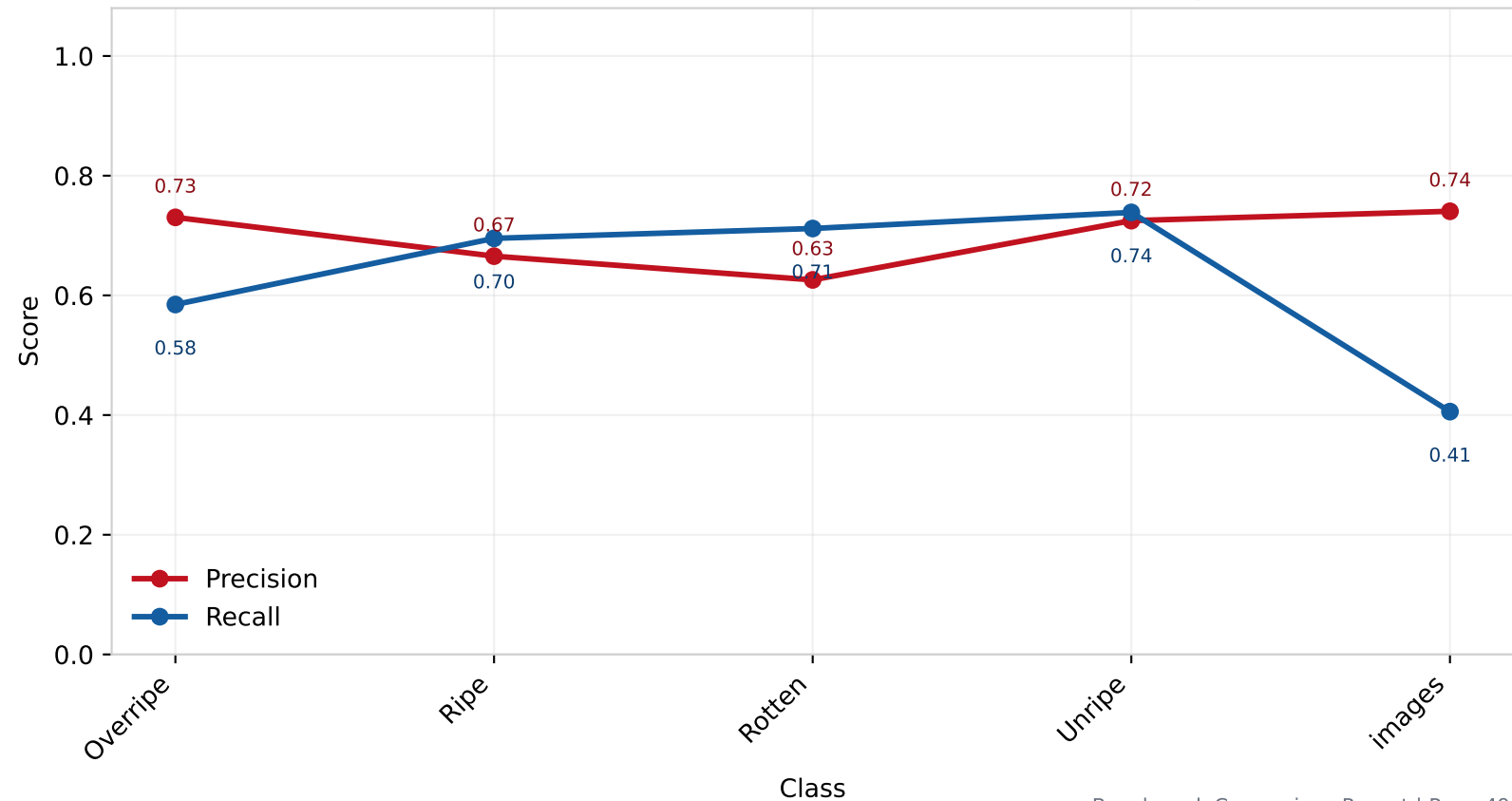
Lower is better

12. Ripeness Classification Prediction Time

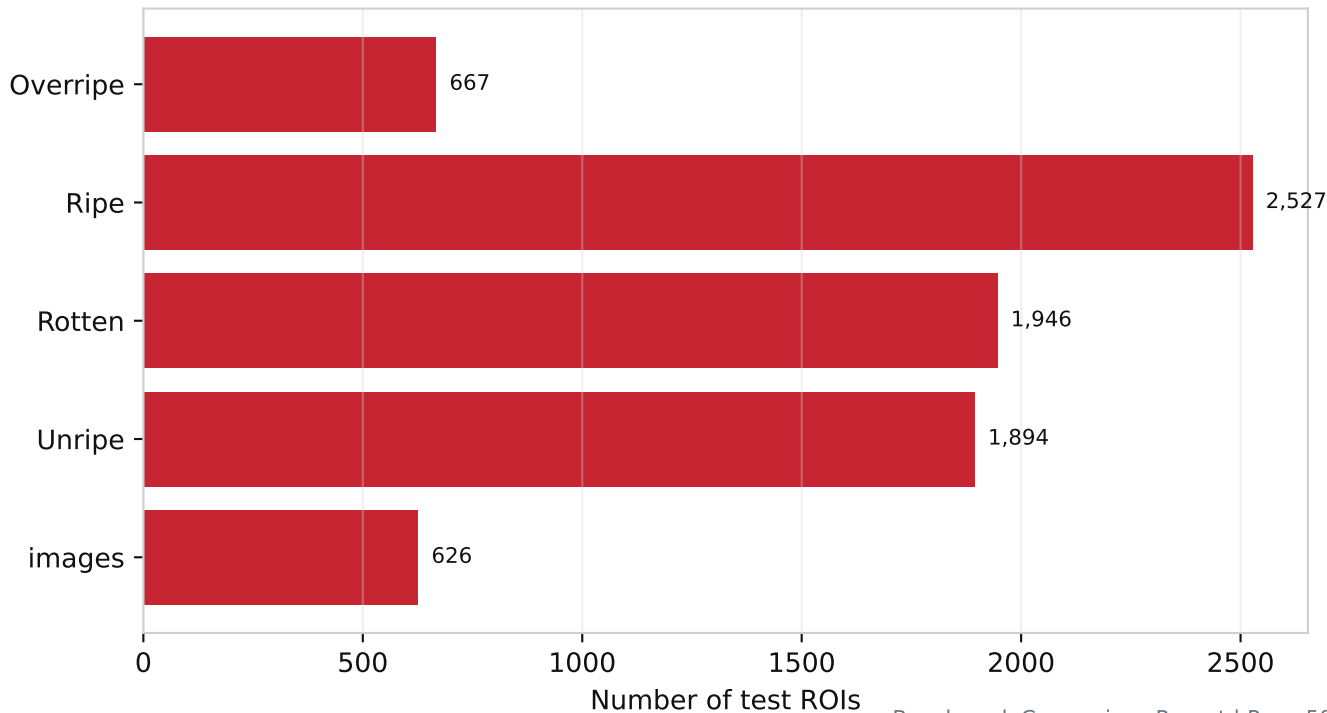


Lower is better

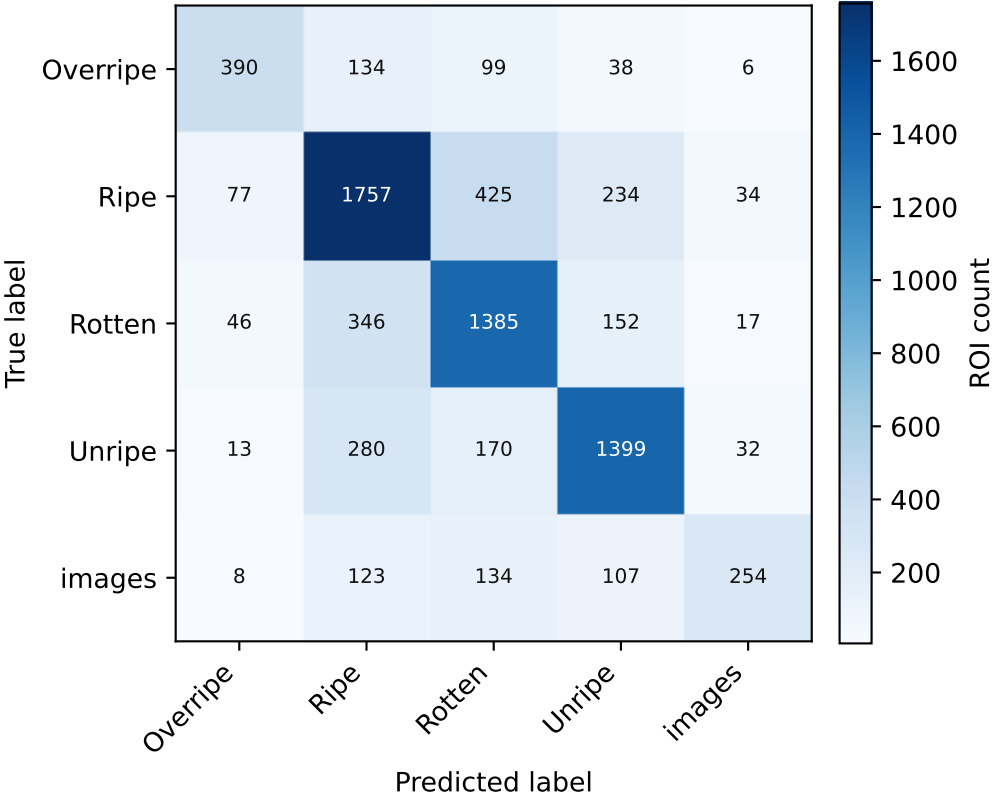
Benchmark A Ripeness — Precision and Recall by Class



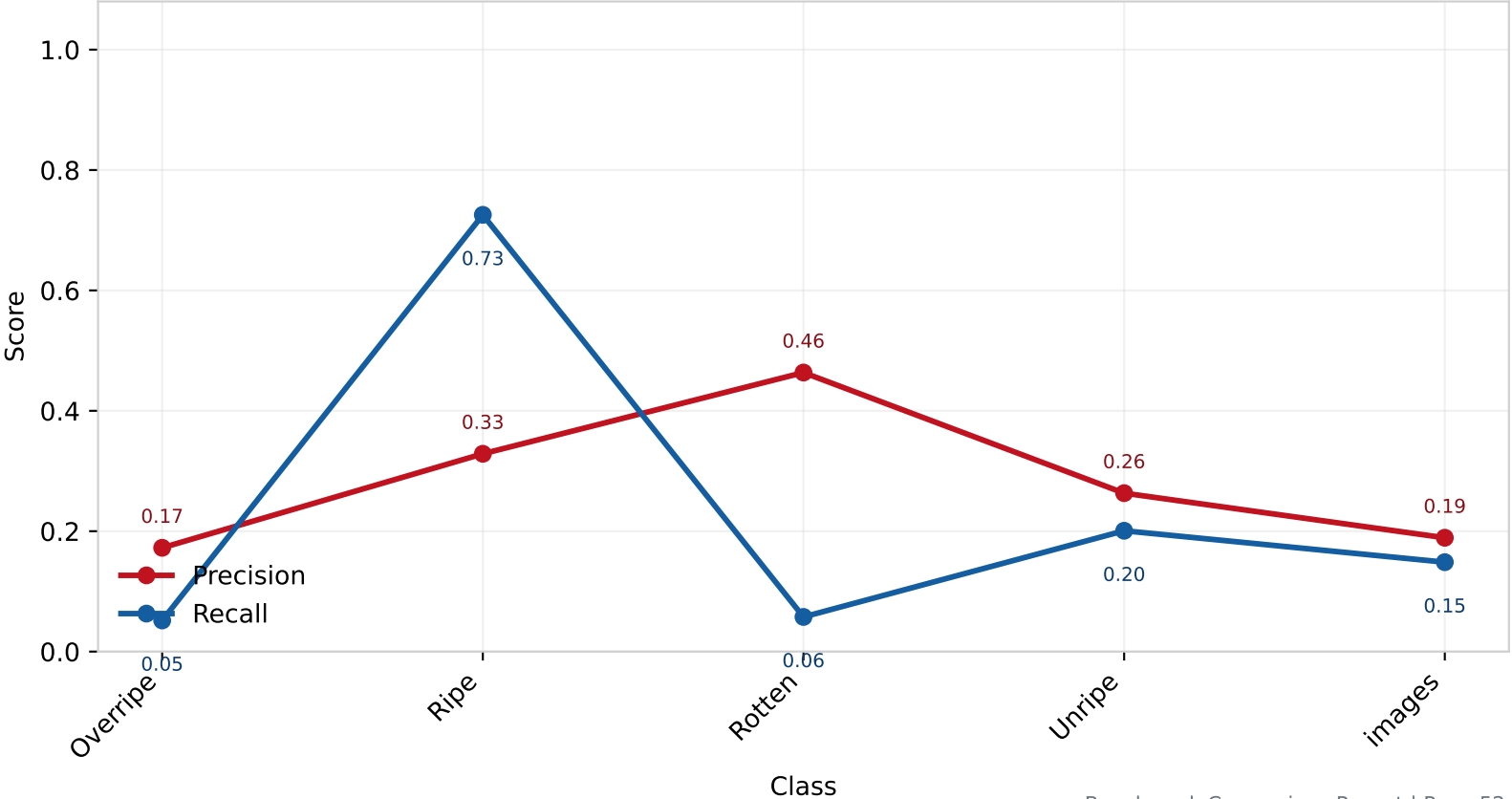
Benchmark A Ripeness Test Support



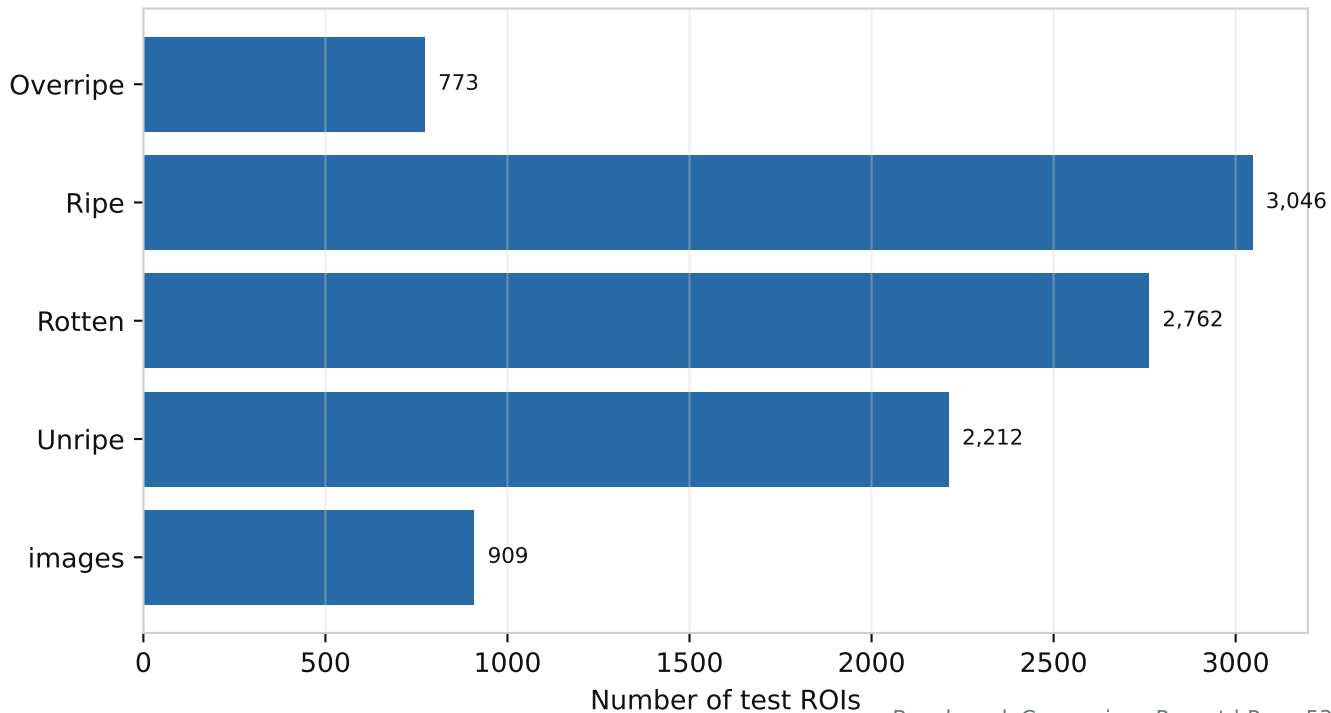
Benchmark A Ripeness Confusion Matrix



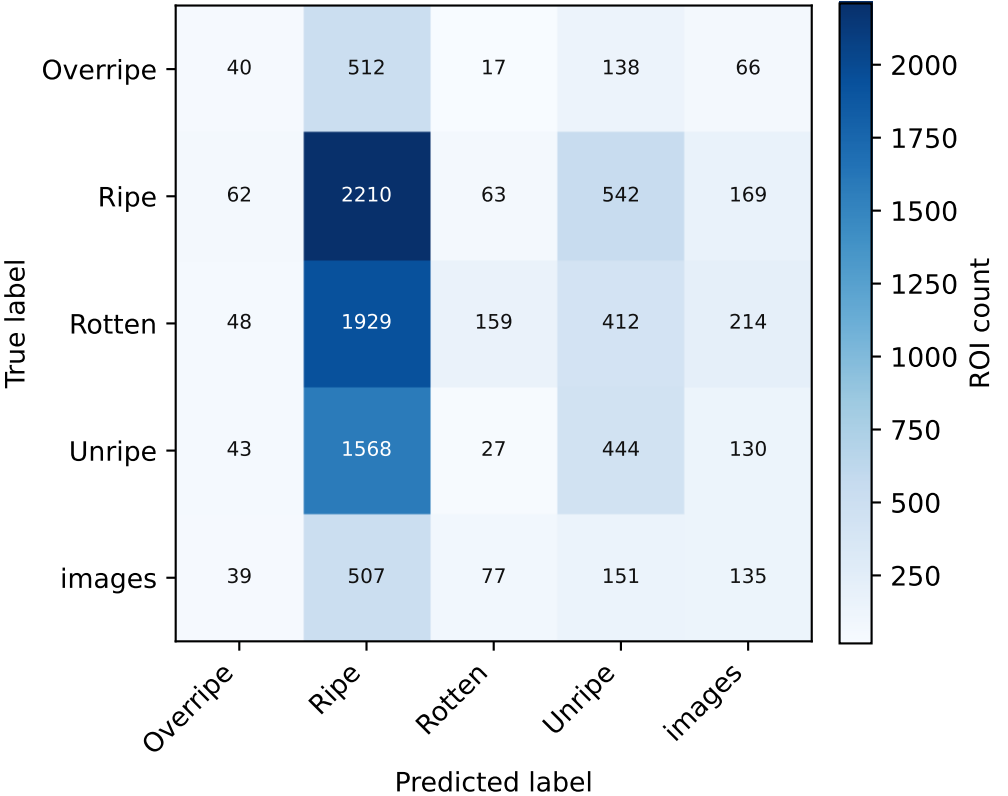
Benchmark B Ripeness — Precision and Recall by Class



Benchmark B Ripeness Test Support



Benchmark B Ripeness Confusion Matrix



Fruit Type Classification Test Results

Fruit type classification is evaluated using Accuracy, Macro Precision, Macro Recall, Macro F1, Training Time and Prediction Time.

Benchmark	Technique	Accuracy	Macro Precisi	Macro Recall	Macro F1	Training Time	Prediction Tir (ms/ROI)	Test ROIs
Benchmark A	LightGBM	0.7544	0.6629	0.5634	0.5894	7.148	0.0364	7,660
Benchmark B	KNN	0.2955	0.1963	0.1856	0.1738	0.153	0.0201	9,702

Ripeness Classification Test Results

Ripeness classification is evaluated using Accuracy, Macro Precision, Macro Recall, Macro F1, Training Time and Prediction Time.

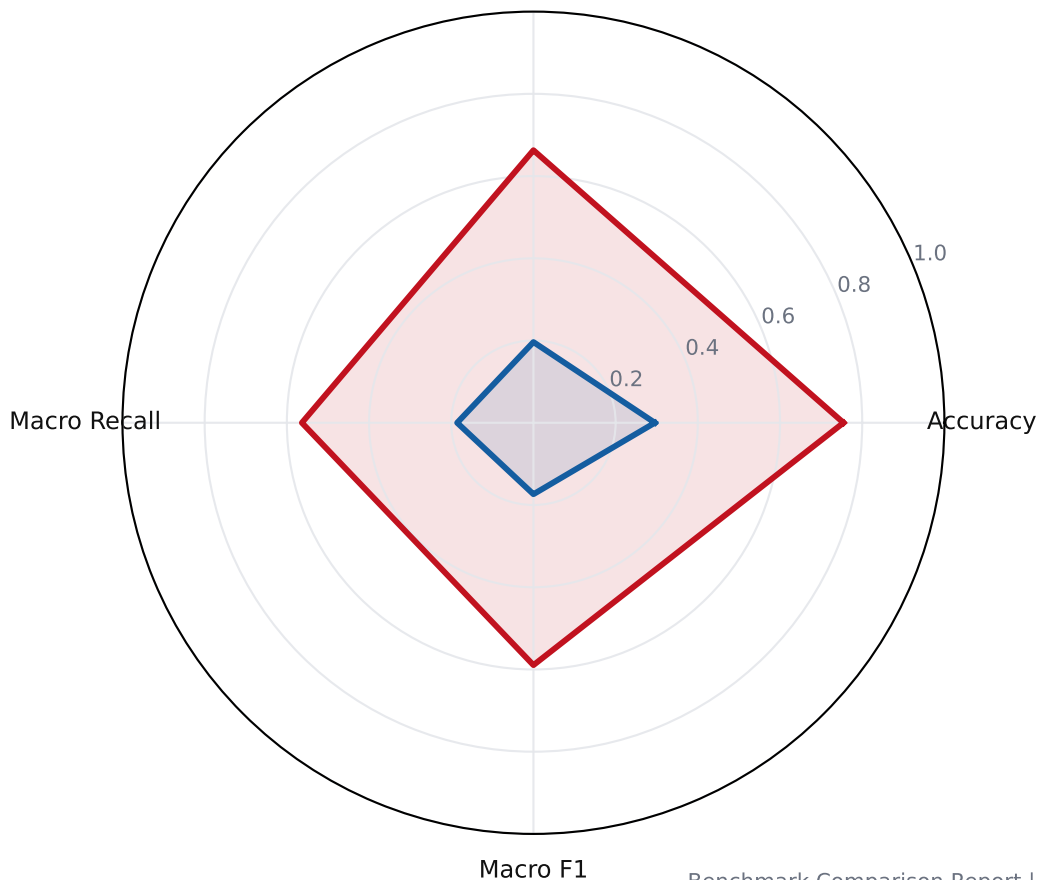
Benchmark	Technique	Accuracy	Macro Precisi	Macro Recall	Macro F1	Training Time	Prediction Tir (ms/ROI)	Test ROIs
Benchmark A	CatBoost	0.6769	0.6974	0.6272	0.6503	30.197	0.0011	7,660
Benchmark B	Decision Tree	0.3080	0.2834	0.2368	0.2057	0.274	0.0004	9,702

Fruit Type Classification Radar

Macro Precision

Benchmark A

Benchmark B

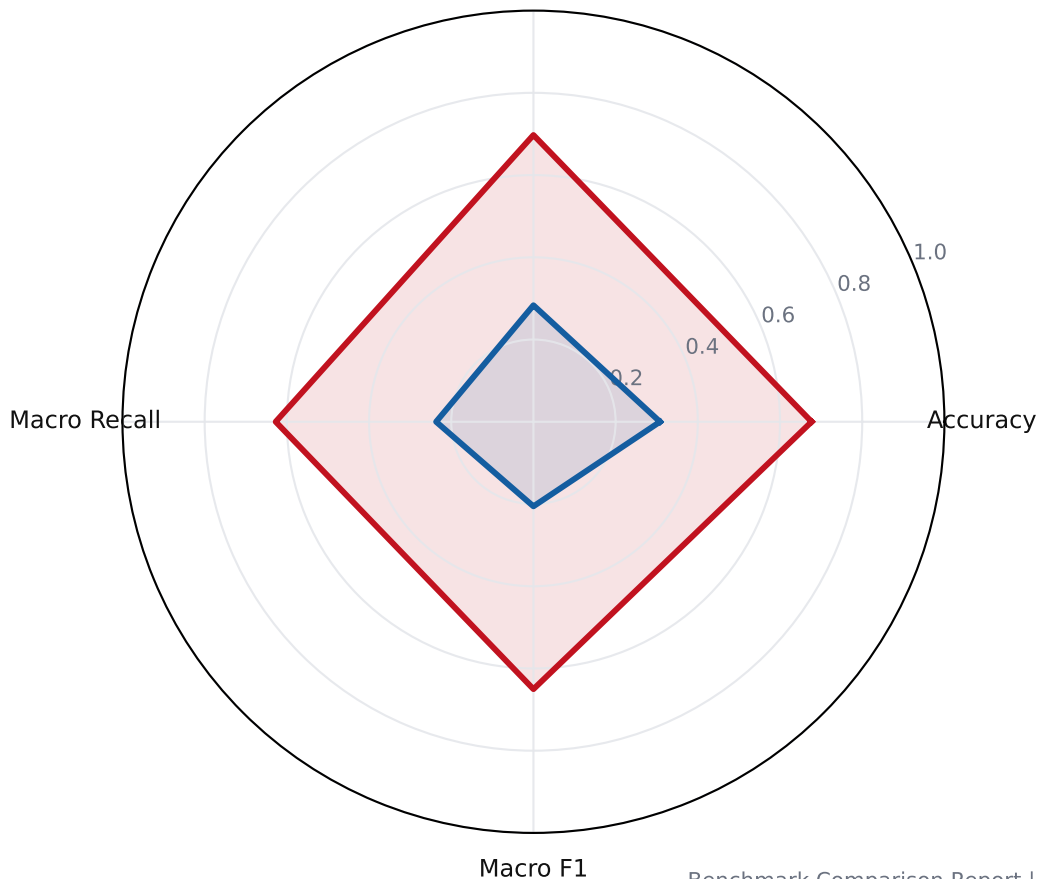


Ripeness Classification Radar

Macro Precision

Benchmark A

Benchmark B



Final Hybrid Pipeline Selection Summary

The Hybrid Pipeline can be assembled using the winning technique from each evaluated stage. The stage score is based only on saved metrics that are available for both benchmarks. Missing metrics are not assigned artificial values.

Overall Stage Winner

Benchmark A

Benchmark A Points 24.0

Benchmark B Points 11.0

Stage	Winner	Benchmark A Score	Benchmark B Score	Metrics Compared
3. Preprocessing	Benchmark A	3.0	2.0	5
4. Segmentation	Benchmark A	3.0	1.0	4
5. Morphological Processing	Benchmark A	3.0	1.0	4
6. Fruit Detection	Tie	1.0	1.0	2
7. Fruit Counting	Benchmark A	4.0	0.0	4
10. Feature Extraction	Tie	2.0	2.0	4
11. Fruit Type Classification	Benchmark A	4.0	2.0	6
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